



Artificial intelligence and adaptive response to market changes: A strategy to enhance firm performance and innovation

Yulia Sullivan^a, Samuel Fosso Wamba^{b,*}

^a Hankamer School of Business, Baylor University, One Bear Place #98005, Foster Campus, Waco, TX, USA

^b TBS Business School, France 1 Place Alphonse Jourdain, 31068 Toulouse, France

ARTICLE INFO

Keywords:

Artificial intelligence
Adaptive response to market changes
Innovation
Firm performance
AI-powered capabilities

ABSTRACT

This research examines how AI-powered capabilities can bring value to organizations by enhancing their Adaptive Response to Market Changes (ARMC). Utilizing insights from organizational agility and the dynamic capability framework, we define ARMC as an organization's ability to promptly identify and adjust to market changes, with customer responsiveness and operational adjustment as foundational competencies. We outline three AI-powered capabilities (AI-enabled automation, AI-enabled analytics, and AI-enabled relational capabilities) as ARMC's predictors. We posit that the strengths of these relationships depend on environmental hostility and dynamism. Additionally, we propose positive associations between ARMC and three organizational outcomes: firm performance, process innovation, and product innovation. Our research employs a two-stage design, surveying IT and business executives from firms that have adopted AI. The results demonstrate significant interaction effects of environmental hostility and dynamism on the relationships between AI-powered capabilities and ARMC. Furthermore, we find that ARMC positively influences firm performance and innovation.

1. Introduction

The rise of digitization, characterized by the profound use of data analytics and artificial intelligence (AI), has compelled organizations to adopt a new perspective on their IT assets and focus on a more intricate level (Benner & Tushman, 2015). Typically defined as “the ability of machines to perform human-like cognitive tasks,” AI is currently viewed as the most important and disruptive new technology for organizations (Benbya et al., 2020, p. ix). However, the management of AI differs significantly from previous approaches to IT management (Berente et al., 2021). AI cannot be viewed merely as a technology or a collection of technologies; rather, it represents an ever-evolving system encompassing emerging computational capabilities (Berente et al., 2021). Thus, today's leaders must navigate both the opportunities and the obstacles that come with the pervasive adoption of AI.

Further, the process of digitalization enabled by AI has triggered a paradigm shift in the fundamental theories that explain how IT innovation generates value (Nambisan et al., 2017). While the challenges of utilizing IT to create business value are fairly well understood (Barua et al., 2004; Côte-Real et al., 2017; Grover et al., 2018; Kohli & Grover, 2008; Melville et al., 2004; Wang et al., 2012), the recent rise of AI

introduces a new set of challenges to this longstanding issue. Consequently, more contemporary scrutiny and analysis is needed (Nambisan et al., 2017). The emergence of new AI capabilities and their associated challenges raises the question of how firms can effectively harness these technological advancements to gain a competitive edge in their industries (Iansati & Lakhari, 2020).

Although prior studies have shown the promising role of AI in supporting decision-making processes in an organization (e.g., Duan et al., 2019; Gupta et al., 2022; Jussupow et al., 2021; Shrestha et al., 2019), theoretical consensus is limited on how AI influences firms' responses and adaptability to market changes. However, AI can empower organizations to respond to dynamic market changes effectively (Narayandas & Sengupta, 2023). For example, it can automate and optimize various operation processes, enhancing the efficiency and responsiveness of an organization (Benbya et al., 2020; Huang & Rust, 2021). AI-enabled predictive analytics allow organizations to identify emerging market trends and connect with customers, allowing them to proactively adapt their strategies and offerings (Huang & Rust, 2021). Moreover, by leveraging AI, businesses can gain valuable insights from vast volumes of data and make informed decisions in real time (Davenport et al., 2020). In sum, AI provides organizations with advanced computing

* Corresponding author.

E-mail addresses: yulia_sullivan@baylor.edu (Y. Sullivan), s.fosso-wamba@tbs-education.fr (S. Fosso Wamba).

<https://doi.org/10.1016/j.jbusres.2024.114500>

Received 10 March 2022; Received in revised form 29 December 2023; Accepted 4 January 2024

Available online 13 January 2024

0148-2963/© 2024 Elsevier Inc. All rights reserved.

abilities to perform mechanical tasks (i.e., *AI-enabled automation capability*), process information and complete analytics tasks (i.e., *AI-enabled analytics capability*), and directly interact with humans (i.e., *AI-enabled relational capability*) (Huang & Rust, 2021a, 2021b). Thus, these computing abilities help companies generate new ways to enhance their responses to market opportunities.

In this study, we explore how AI-powered capabilities can create value for organizations, leading to enhanced firm performance and innovation. These capabilities are crucial in establishing firms' ability to swiftly identify and respond to market changes, such as shifts in customer preferences or new market trends. We refer to this concept as "Adaptive Response to Market Changes" (ARMC). Our emphasis is on adopting a core competencies perspective (Ravichandran et al., 2005) to effectively reflect how firms respond to market changes in two key areas: customer responsiveness and operational adjustment.

Furthermore, environmental conditions influence firms' capability-building processes and responses to market demands (Lee et al., 2015). However, limited research has explored how these conditions affect firms' responses to market changes concerning AI even though they play a crucial role in shaping managerial decision-making (Pavlou & El Sawy, 2011; Wilden & Gudergan, 2015). While strategic decisions in product or market adaptation can be unpredictable in rapidly changing environments (Eisenhardt & Martin, 2000; Mitchell et al., 2011), our study proposes that the effects of AI-powered capabilities on ARMC depend on specific environmental conditions. Specifically, we theorize that two environmental conditions—*environmental hostility* and *environmental dynamism*—affect the degree to which AI-powered capabilities drive firms' responses to market changes.

Focusing on this objective, we will elucidate how the interplay between AI-powered capabilities and environmental conditions can result in different levels of ARMC, which directly affects firm performance, process innovation, and product innovation. Our study seeks a deeper understanding of the circumstances under which AI-powered capabilities will likely yield benefits or drawbacks for organizations. Moreover, our investigation into how AI-powered capabilities facilitate ARMC contributes to AI research beyond its purely technical aspects (e.g., Asatiani et al., 2020). Thus, the empirical insights from this study offer practical guidance to organizations, enabling them to anticipate and prepare for the economic impacts of AI implementation.

2. Theoretical background

2.1. An overview of dynamic capabilities

Dynamic capabilities are defined as "a firm's ability to integrate, build, and reconfigure internal and external competencies to address rapidly changing environments" (Teece et al., 1997, p. 516). Dynamic capabilities are distinct from operational capabilities because they serve different organizational functions. While operational capabilities are essential for companies' smooth functioning in day-to-day operations, dynamic capabilities help organizations achieve and sustain a competitive advantage (Lin & Kunnathur, 2019; Teece, 2007; Teece et al., 1997). Firms demonstrate dynamism in capabilities by adapting strategies to changing market conditions and creatively combining existing resources in innovative and unconventional ways (Morgan et al., 2009).

Dynamic capabilities constitute firms' ability to (1) sense and shape opportunities and threats, (2) respond to market opportunities, and (3) maintain competitiveness through enhancing, aligning, and reconfiguring firms' intangible and tangible resources (Teece et al., 1997). Although the primary idea of dynamic capabilities centers on these sensing, seizing, and reconfiguring activities (Teece, 2007), scholars have proposed several other dynamic capabilities, including dynamic managerial capabilities (Helfat & Peteraf, 2015), organizational agility (Chakravarty et al., 2013; Nazir & Pinsonneault, 2012; Sambamurthy et al., 2003), market orientation (Morgan et al., 2009), and new product development (Pavlou & El Sawy, 2006).

In the context of ARMC, organizational agility emerges as the most closely related concept, encompassing companies' ability to flexibly adapt to shifting market conditions and seize opportunities for sustainable growth and competitiveness. In their recent work, Kalaiganam et al. (2021) explored the commonalities and distinctions among dynamic capabilities, agility, and associated concepts. According to their findings, dynamic capabilities encompass *sensemaking*, which involves firms' ability to respond to unexpected events by discerning the events unfolding and striving to establish a more structured environment, and *speed*, which represents the swiftness with which firms sense changes and initiate responses. However, dynamic capabilities lack the element of iteration. In contrast, agility includes the critical aspect of *iteration*, entailing the iterative refinement of decisions before their relaunch or scaling to minimize uncertainty. Thus, while agility and dynamic capabilities overlap, they also have distinct features, such as agility emphasizing the iterative nature of decision-making to reduce uncertainty.

The dynamic capabilities framework also recognizes the importance of agility. However, it posits that the continual pursuit of agility may not always be necessary since doing so could lead to a loss of efficiency and additional costs. Instead, agility becomes imperative primarily when market changes are linked to uncertainty or risks (Teece et al., 2016). Put differently, firms may not require constant agility except in situations where uncertainty is present. Consequently, when firms encounter uncertainty, they can enhance their agility by leveraging dynamic capabilities. Next, we explain ARMC from the agility and dynamic capabilities framework.

2.2. Adaptive response to market changes (ARMC)

We derive the concept of ARMC from the organizational agility perspective. Scholars have offered various definitions of agility depending on the context. For example, in the marketing field, Kalaiganam et al. (2021) conceptualized marketing agility as "the extent to which an entity rapidly iterates between making sense of the market and executing marketing decisions to adapt to the market" (p. 36). Meanwhile, in the information systems (IS) field, Tallon and Pinsonneault (2011) conceptualized organizational agility as "the ability to detect and respond to opportunities and threats with ease, speed, and dexterity" (p. 464). Sambamurthy et al. (2003) also proposed that agility comprises three interrelated capabilities: customer, partnering, and operational agility. Moreover, Lu and Ramamurthy (2011) conceptualized two dimensions of organization agility: market capitalizing and operational adjustment agility.

Organizational agility, as highlighted by Lu and Ramamurthy (2011), represents a comprehensive firm-wide capability to effectively navigate through unexpected changes in dynamic business environments. Given the breadth of this concept, encompassing all its elements within a single study is challenging. Nonetheless, in our research, we focus on two crucial elements related to organizational agility that ARMC¹ captures: *customer responsiveness* and *operational adjustment/adaptiveness* dimensions.² These two dimensions specifically focus on the two core competencies found within organizations, namely market-

¹ Instead of utilizing the term "organizational agility," which encompasses a firm-wide capability, we opt for the more precise term Adaptive Response to Market Changes (ARMC) to refer to the specific focus of our study. We thank the Handling Editor for this suggestion.

² Utilizing Kalaiganam et al.'s (2021) criteria for agility, we contend that customer responsiveness entails marketing decisions, especially those related to customer or product choices, in addition to sensemaking and speed. On the other hand, operational adjustment involves sensemaking and speed. Although we acknowledge that the iteration component of agility is difficult to assess in a cross-sectional study, we assume that for decision-makers to arrive at market decisions, they must adapt and refine the problem.

access and functional-related competencies (Battleson et al., 2016; Bradley et al., 2012; Chen et al., 2017).

Market-access competencies encompass all activities that enable firms to connect with their customers and effectively identify their needs (Ravichandran et al., 2005). In the context of ARMC, customer responsiveness represents a crucial aspect of market-access competencies, encompassing companies' ability to actively engage with customers to foster collaboration, explore innovative opportunities, and develop competitive strategies in response to market changes (Sambamurthy et al., 2003). For instance, customer responsiveness could include adapting to shifts in aggregate customer demand and customizing products or services to cater to individual customer preferences. Operational-related competencies encompass all internal business process activities that enable firms to deliver distinctive and dependable products while enhancing the speed and precision of key operations (Chen et al., 2017). Within these competencies, operational adjustment assumes significant importance. Operational adjustment refers to firms' ability to promptly modify their internal business processes, facilitating effective adaptation to shifts in market demands (Lu & Ramamurthy, 2011). For instance, operational adjustments could involve expanding into a new market and adjusting the range of products or services offered for sale.

2.3. Artificial intelligence and AI-Powered capabilities

AI is "the frontier of computational advancements that references human intelligence in addressing ever more complex decision-making problems" (Berente et al., 2021, p. 1435). Currently viewed as the most important and disruptive new technology for organizations (Benbya et al., 2021), AI is revolutionizing managerial practices for analyzing performance and developing novel insights (Kaplan & Haenlein, 2019). In addition, it includes statistical machine learning, neural networks, and natural language processing (NLP) (Davenport & Mahidhar, 2018).

Technology is widely acknowledged as a fundamental resource that organizations can effectively exploit, deploy, and leverage to their advantage (Manis & Madhavaram, 2023). In light of the capability-building process, the significance of investments in IT lies in its successful integration into the core activities of organizations (Doherty & Terry, 2009; Ravichandran et al., 2005; Sambamurthy et al., 2003). In this context, Information Systems (IS) serves as a crucial mechanism in establishing coherence between IT activities and firm priorities, thereby facilitating the development of comprehensive capabilities that enable seamless cross-functional integration and coordination (Chen et al., 2017; Ravichandran et al., 2005). Through this lens, our focus centers on three AI-powered capabilities that are related to three important business needs: automating business processes, gaining insights, and engaging with customers (Davenport & Ronanki, 2018).

These AI-powered capabilities include (1) *AI-enabled automation capability*, also referred to as mechanical AI; (2) *AI-enabled analytics capability*, known as cognitive insights or thinking AI; and (3) *AI-enabled relational capability*, also known as feeling AI or cognitive engagement (Davenport & Ronanki, 2018; Huang et al., 2019; Huang & Rust, 2021a). Although these three AI capabilities are interrelated, each supports business functions differently (Huang & Rust, 2021b). Each AI-powered capability brings unique advantages: AI-enabled automation excels in standardization, AI-enabled analytics capability is most effective for personalization, and AI-enabled relational capability proves ideal for fostering strong relationships (Huang & Rust, 2021a, 2021b). The definition of each AI-powered capability and examples of each are summarized in Table 1.

The first AI capability—*AI-enabled automation capability*—is used to automate repetitive and routine tasks (Huang & Rust, 2021b; Makarius et al., 2020). AI-enabled automation is often used to improve output consistency, standardization, reliability, and precision (Huang et al., 2019). The second capability—*AI-enabled analytics capability*—can be

Table 1
Conceptualization of AI-powered capabilities.

AI Capability	Definition	Examples
AI-enabled automation capability	AI's capability to do mechanical and repetitive tasks with relatively limited amounts of learning or adaptation (Huang et al., 2019)	Using AI to get information; using AI to perform general physical activities; using AI to perform administrative activities (Huang et al., 2019).
AI-enabled analytics capability	AI's capability to analyze and make decisions based on vast volumes of data (Davenport & Ronanki, 2018)	Using AI to personalize products based on customers' preferences; using AI to anticipate credit card fraud; using AI to predict equipment failures (Davenport & Ronanki, 2018).
AI-enabled relational capability	AI's capability to recognize, emulate, and respond appropriately to human requests by learning and adapting from experience (Huang et al., 2019)	Using AI bots to provide sales and customer service supports; using virtual agents to assist and care for others (Huang et al., 2019).

described as "analytics on steroids" (Davenport & Ronanki, 2018, p. 4). AI with this capability generally uses machine-learning algorithms to detect patterns and interpret their meanings in decision-making processes (e.g., predict what a particular segment of customers is likely to buy and analyze warranty data to identify the safety problems in automobiles and other manufactured products) (Huang et al., 2019). Unlike traditional analytics, AI-enabled analytics involves learning and adapting systematically from data (Huang et al., 2019). An AI system can process both unstructured and structured data, as well as information generated by the system itself to acquire new or modify existing knowledge (Paschen et al., 2020). Lastly, *AI-enabled relational capability* engages with human users using NLP chatbots, intelligent agents, and machine learning (Davenport & Ronanki, 2018). AI with a relational capability allows users to directly engage with a virtual agent rather than with human customer-service agents (Davenport & Ronanki, 2018). Such AI can learn and adapt its behaviors by learning from its interactions with human users (Huang & Rust, 2021a).

3. Research model and hypotheses development

3.1. Research model

Fig. 1 presents our conceptual model. In proposing our research model, we posit that the three AI-powered capabilities act as predictors of ARMC, which measures how quickly and easily a firm can respond to opportunities within two areas: customer responsiveness and operational adjustment. According to the underlying theory (e.g., Sambamurthy et al., 2003), the first-order dimensions of ARMC are complementary (i.e., they interact and co-vary with each other). Thus, following guidelines provided by Tanriverdi and Venkatraman (2005), we conceptualize ARMC as a reflective, second-order construct consisting of two first-order dimensions: customer responsiveness and operational adjustment. Furthermore, considering that tasks can differ in their degree of automation, analytics, and relational intelligence capability (e.g., information acquisition often involves more mechanical processes, whereas information processing and analysis are more analytics-driven) (Huang et al., 2019), we perceive the three AI-powered capabilities as interrelated yet distinct constructs. Thus, we conceptualize them as reflective, first-order constructs.

Furthermore, we consider the impact of environmental conditions as we explore the relationships between AI-powered capabilities and ARMC. We hypothesize that the effects of AI-powered capabilities on ARMC may vary depending on the environmental conditions. Specifically, we aim to comprehend the influence of perceived hostile and dynamic environmental contexts (Mitchell et al., 2011). Finally, we

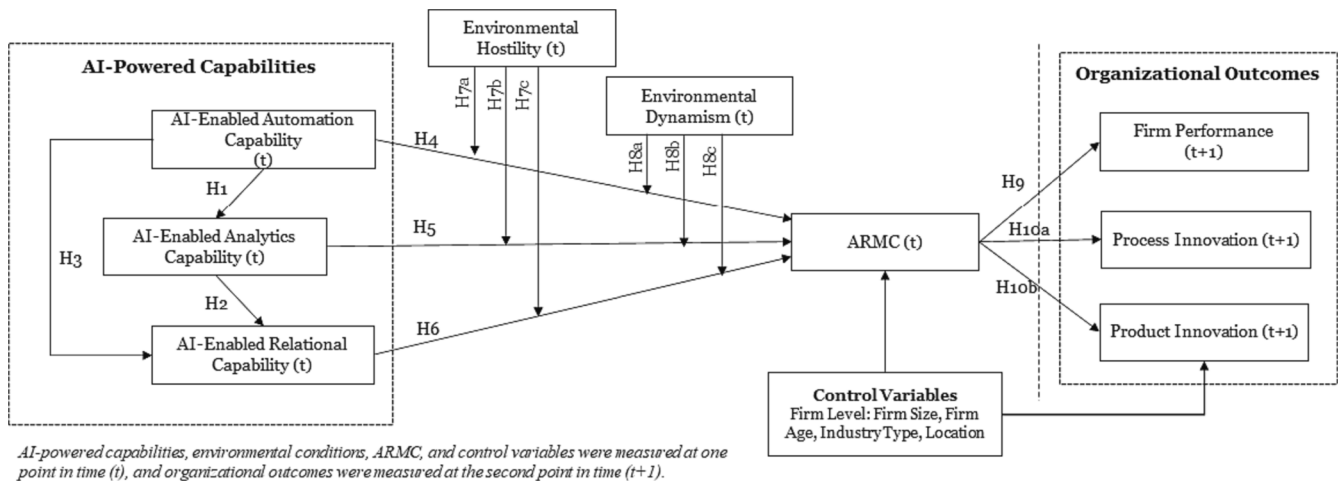


Fig. 1. Conceptual model.

establish a connection between ARMC and our primary outcomes of interest: firm performance, process innovation, and product innovation. In our investigation, we adopt an agility perspective to view firm performance as an outcome of ARMC (Chen et al., 2014; Holmqvist & Pessi, 2006; Kathuria et al., 2018). Additionally, we employ the dynamic capabilities perspective, which suggests that firms with strong sensing and adaptive response abilities can drive innovation and new product development (Hoonsopon and Puriwat, 2019; Puriwat and Hoonsopon, 2021; Teece et al., 2016). Thus, we also measure process and product innovation as our dependent variables.

3.2. Hypotheses development

3.2.1. The relationships among AI-Powered capabilities

According to Huang and Rust's (2021a) perspective of multiple AIs, the advancement of AI occurs cumulatively. In this context, AI progresses through higher levels while retaining its lower-level capabilities. From this perspective, AI-enabled automation represents the lowest and simplest level, followed by AI-enabled analytics capability, and, finally, AI-enabled relational capability. Therefore, we assert that AI-enabled automation serves as the fundamental building block for firms' analytics and relational capabilities. AI-enabled automation empowers firms to perform repetitive tasks, albeit with limited learning and adaptive capabilities (Huang et al., 2019). Considering that this type of AI is the most cost-effective and easiest to implement (Davenport & Ronanki, 2018), we can reasonably assume that firms would typically prioritize establishing this AI capability as their initial step. Once this capability has been established, they can utilize it to enhance AI-enabled analytics capability through algorithms and models as a means of learning, as well as AI-enabled relational capability to establish meaningful relationships with users ((Huang & Rust, 2022)).

Further, AI-enabled analytics capability can significantly enhance firms' relational capability, leading to improved business decision-making, especially in the context of future collaborations with human users (Huang & Rust, 2022). This result becomes evident when AI-powered analytics efficiently uncover valuable insights and personalize user interactions (Davenport et al., 2020). Furthermore, as AI continually acquires new insights and learns through its algorithms, it becomes increasingly adept at recognizing and responding appropriately during human interactions (Huang et al., 2019). Thus, our first three hypotheses concern the relationships among AI-powered capabilities:

H1: AI-enabled automation capability is positively associated with AI-enabled analytics capability.

H2: AI-enabled analytics capability is positively associated with AI-enabled relational capability.

H3: AI-enabled automation capability is positively associated with AI-enabled relational capability.

3.2.2. AI-enabled automation capability and ARMC

AI-enabled automation capability plays a vital role in enhancing ARMC by automating routine tasks and enabling team members to focus on more impactful high-level activities for the business (Raju & Koch, 2019). A notable example is Crédit Agricole Technologies et Services (CA-TS), which employs Robotic Process Automation (RPA) to automate its project validation process, resulting in faster decision-making (Deloitte, 2020). AI-enabled automation not only breaks down organizational siloes by automating data sensing and collection processes, but also delivers standardized benefits through its consistency (e.g., using robots for packaging tasks) (Huang & Rust, 2021b). By eliminating unnecessary traditional data collection methods, automation capability helps firms respond to market changes by reducing redundancies and inefficiencies (Huang et al., 2019).

The strategic deployment of AI-enabled automation is also considered a contingency-driven decision, empowering organizations to respond to market changes by automating routine and time-consuming processes, thereby reducing production costs and risks associated with delivering products/services. Thus, AI-enabled automation can improve scalability (Raju & Koch, 2019). In turn, firms can efficiently redeploy/redirect their resources to value creating and value capturing activities (Teece et al., 2016). Therefore, we hypothesize the following:

H4: AI-enabled automation is positively associated with ARMC.

3.2.3. AI-enabled analytics capability and ARMC

The relationship between AI-enabled analytics capability and ARMC can be explained through the lens of sensing (Teece et al., 2016). In environments characterized by uncertainty, companies must sense and generate digital options for growth before the market logic of these options becomes apparent to all (Teece et al., 2016). AI functionalities, such as predictive analytics, diverse customer data analysis, and personalized content creation, provide a range of digital options for organizations. These options enhance process efficiency, knowledge alternatives, and rapid response to changes. For instance, from a market-access perspective, integrating AI into products and services allows companies to generate and test models swiftly, enhancing the ability to explore new markets, develop innovative products, and expedite product development (Davenport, 2018). From an operational-related standpoint, AI analytics capability offers novel problem-solving approaches and supports human decision-making by overcoming inherent information-processing limitations (Haefner et al., 2021). Thus, AI-enabled analytics capability is crucial in facilitating ARMC, leading to the personalization of products and services by supporting decision-

making processes throughout the entire value chain.

According to the dynamic capability perspective, undertaking actions to proactively create hypotheses about the future implications of observed events and trends can help organizations sense and seize market opportunities (Teece et al., 2016). For example, firms can use machine learning to make predictions and assess their quality in practice (Albrecht et al., 2021; Capatina et al., 2020; Davenport et al., 2020). They can also track users' actions and responses in the field, offering accurate suggestions that capture users' increasing interest over time (Hollebeek et al., 2021). Since AI analytics tools provide data-driven knowledge (Kiron & Schrage, 2019), they offer accurate and reliable information, enabling firms to flexibly respond to market opportunities that might be overlooked by humans operating alone (Haefner et al., 2021). Thus, we hypothesize the following:

H5: AI-enabled analytics capability is positively associated with ARMC.

3.2.4. AI-enabled relational capability and ARMC

AI-enabled relational capability goes beyond merely detecting patterns in data and providing insights for decision-making processes; it involves using data to train AI systems that can interact with users without human intervention (Davenport & Ronanki, 2018). AI's relational capability is used to engage users in NLP chatbots, deep learning, and machine learning (Duan et al., 2019). When employed in user interaction, an AI application with a relational capability offers several business benefits: it facilitates firm-initiated communications, engages with customers in a friendly manner, is unaffected by frustration or fatigue like humans are, and can easily scale up to handle a large volume of customer communications (Luo et al., 2019). These advantages empower firms to listen attentively to users and address their concerns effectively.

The relationship between relational capability and ARMC can also be explained from the data network effects perspective (Gregory et al., 2021). A platform exhibits data network effects if the more it learns from the data it collects on users, the more valuable the platform becomes to each user. For example, Uber uses machine-learning algorithms to analyze users' data and improve its algorithmic matching, information, and experience offered to users who engage in exchanges among riders and drivers. Uber's goal is to inform and engage users with push notifications and messages that provide intelligent recommendations that adapt to changing contextual and situational circumstances (Gregory et al., 2021; Rosenblat, 2018). One of the strategic roles of AI in today's business is to continuously improve enterprises' products and services for each user (Gregory et al., 2021), allowing firms to cope with rapid, relentless, and uncertain changes. Therefore, we hypothesize the following:

H6: AI-enabled relational capability is positively associated with ARMC.

3.2.5. The moderating role of environmental hostility and environmental dynamism

Prior research has shown that external environmental factors are important contextual variables in influencing decision-making processes and organizational agility (Chatterjee et al., 2023; Helfat, 2022; Lee et al., 2015; Mitchell et al., 2011; Teece et al., 2016). These factors can significantly impact companies' strategic choices, resource allocation, and overall responsiveness to changing market conditions, thereby playing a crucial role in shaping organizations' ability to adapt to and thrive in a dynamic business landscape (Helfat et al., 2007). In this study, we seek to understand the effect of two environmental contexts (environmental hostility and environmental dynamism) in moderating the relationship between AI-powered capabilities and ARMC.

3.2.5.1. Hostile environmental context. Environmental hostility (Özsomer et al., 1997) indicates "unfavorable external forces for a firm's

business" (Zahra & Garvis, 2000, p. 475). These unfavorable environmental conditions can result from radical industry changes, intense regulatory burdens placed on the industry, or fierce rivalry among competitors (Chen et al., 2014; Zahra & Garvis, 2000). Hostility also results from perceived competitive-, market-, and product-related uncertainties (Dess & Beard, 1984; Zahra & Garvis, 2000). Under hostile environmental conditions, market structures are fluid, clusters of know-how and technologies are globally dispersed, and innovations are dependent on technology combinations (Teece, 2014). Thus, environmental hostility requires firms to break their routines and continuously learn new complexities and uncertainties, leading to increased chances of survival and success in such challenging settings (Teece et al., 2016). Not surprisingly, firms need a strategic direction to ensure survival in such environments (Slevin & Covin, 1997).

Given that AI-enabled automation is primarily used to automate routine tasks (Huang et al., 2019), maximize efficiency, and minimize variability (Huang & Rust, 2021a), firms relying on this capability while operating in highly hostile environments cannot quickly respond to market changes. While AI-enabled automation enhances operational efficiency and reduces human error (Huang & Rust, 2021a), its limitations lie in its inability to adapt quickly to dynamic and unpredictable market conditions (Tussyadiah, 2020). Thus,

H7a: Environmental hostility negatively moderates the relationship between AI-enabled automation capability and ARMC.

In a hostile environment, accurate information is difficult to obtain (Bourgeois & Eisenhardt, 1988), making strategic decision-making difficult due to the uncertainty surrounding the potential significance of certain changes (Özsomer et al., 1997). With AI-enabled analytics capability, firms can anticipate and adapt to market trends (Allal-Chérif et al., 2021). They can also create hypotheses about the future implications of observed events and test these hypotheses to generate pathways for new products or services (Teece et al., 2016). Consequently, in a highly hostile environment, prioritizing AI-enabled analytics capability becomes imperative for achieving ARMC. Thus, we expect the following:

H7b: Environmental hostility positively moderates the relationship between AI-enabled analytics capability and ARMC.

Hostile environments are characterized by severe competitive intensity exacerbated by price wars and customers who exhibit little loyalty to their suppliers (Slevin & Covin, 1997). Among the three AI-powered capabilities, AI-enabled relational capability is the most complex to develop, and many AI applications with this capability are still in the experimental stage (Huang et al., 2019). With its current progress in the industry, developing AI with relational capability may subject firms to higher risks in terms of resource allocation compared to developing AI for analytics purposes. Given that hostile environments are unforgiving (Slevin & Covin, 1997), organizations operating in such conditions should leave little to chance and carefully consider the trade-offs involved in investing resources involving AI with relational capabilities.

Additionally, in a hostile environment, the range of viable solutions for any given organization will likely be restricted such that multiple paths will not be equally workable solutions (Slevin & Covin, 1997). When AI with a relational capability is deployed in a hostile environment, organizations put their relationship with their customers at risk since even the slightest mistake could drive them away, restricting firms' ability to remain flexible. For example, customers may feel uncomfortable talking to AI for personal needs or seeking chatbot assistance in purchase decisions. Humans may also harbor prejudice, perceiving chatbots as lacking personal feelings and empathy, leading to doubts about the bots' trustworthiness with personal information (Luo et al., 2019). Consequently, utilizing AI with a relational capability may potentially decrease firms' ability to respond to the market when they operate in hostile environments. Thus, we hypothesize the following:

H7c: Environmental hostility negatively moderates the relationship between AI-enabled relational capability and ARMC.

3.2.5.2. Dynamic environmental context. A dynamic environment is “one with a highly unpredictable and unstable rate of change and high levels of uncertainty about the state of the context, the means-ends relations, and/or the outcomes of actions” (Mitchell et al., 2011, pp. 687–688). These changes include customer and technology changes (Im & Workman, 2004; Jaworski & Kohli, 1993; Kohli & Jaworski, 1990). For example, the changes brought about by the driving force of technology have created hypercompetitive environments (Kyläheiko & Majanen, 2020; Teece, 2014). In such environments, customers’ demands and competitors’ actions are unpredictable, which requires frequent changes in how businesses operate (Allal-Chérif et al., 2021).

Given that AI-enabled automation learns and adapts to only a minimal degree, it is ideal for service standardization (Huang & Rust, 2021b). For instance, in a low level of dynamism where customers exhibit homogeneous demand or when companies handle routine tasks, AI-enabled automation offers a convenient and efficient operational solution (Huang & Rust, 2021b). However, environmental dynamism requires firms to respond early and rapidly to market changes (Elazhary et al., 2023; Lee et al., 2015). Under such dynamism, AI-enabled automation will be a bottleneck. Its rigid and standardized processes may not sufficiently cater to the diverse and fluctuating demands of customers or adapt to unexpected shifts in the competitive landscape. Thus, when an environment is highly dynamic, overreliance on AI-enabled automation could lead to missed signals and inadequate decision-making, leaving organizations vulnerable to market disruptions. Thus,

H8a: Environmental dynamism negatively moderates the relationship between AI-enabled automation capability and ARMC.

AI-enabled analytics capability allows firms to create new information and predictions from data, even when the data are unstructured (Shrestha et al., 2019). This capability is akin to the generative-sensing capability (Teece et al., 2016) that empowers firms to proactively sense and learn from their environment. By harnessing AI-enabled analytics, organizations can gain valuable insights from vast and complex datasets, enabling them to identify emerging trends, anticipate market shifts, and uncover patterns that might have otherwise remained undetected (Huang & Rust, 2021b; Teece et al., 2016). Possessing this capability provides firms with flexibility and options that they can use to customize and configure their resources (e.g., modify their market segments when needed) as they operate in a dynamic environment (Teece et al., 2016). Thus, environmental dynamism positively moderates the relationship between AI-enabled analytics and ARMC because it amplifies the importance of adaptability, flexibility, and data-driven insights within organizations:

H8b: Environmental dynamism positively moderates the relationship between AI-enabled analytics capability and ARMC.

With a high environmental dynamism, firms face significant unpredictability regarding their customers’ tastes and preferences (Lee et al., 2015). Thus, their IT initiatives should respond to emerging and fast-changing business needs (Chakravarthy et al., 2013). Unlike in a hostile environment where customers are unlikely to exhibit loyalty to firms, customers in a dynamic environment tend to seek adaptability, responsiveness, and personalized experiences. In such an environment, AI-enabled relational capability may positively impact ARMC because it allows firms to nurture long-lasting relationships and foster customer loyalty (Huang & Rust, 2021a). The relational capability allows firms to continuously monitor and quickly improve product or service offerings to address customer needs (Lu & Ramamurthy, 2011). Relatedly, being responsive to customers’ requests requires simultaneous adjustments to firms’ internal business processes (Teece et al., 2016). Using AI to enhance customer relations can provide organizations with better opportunities to respond to market demands and flexibly rearrange their internal business processes. Thus,

H8c: Environmental dynamism positively moderates the relationship between AI-enabled relational capability and ARMC.

In conclusion, the impact of environmental factors on the relationship between AI-powered capabilities and ARMC vary based on

contingency effects. These effects are observed as fluctuations in various components of the external environment, including customers, competitors, and technology (Chakravarty et al., 2013). We have summarized the interaction effects of environmental conditions in Table 2.

3.2.6. ARMC and firm performance

We examine the relationship between ARMC and firm performance by drawing upon the agility literature (e.g., Chen et al., 2014; Kathuria et al., 2018; Ngai et al., 2011; Queiroz et al., 2018; Ravichandran, 2018). In this study, we define firm performance in terms of market and financial performance. ARMC can improve performance by expanding firms’ range of competitive actions and the nature of their feasible responses to environmental change (Tallon & Pinsonneault, 2011). Firms with high ARMC can take advantage of market opportunities to improve delivery performance, reduce reaction time (Anand & Ward, 2004), and, in turn, achieve high performance (Chen et al., 2017). Firms with high ARMC, therefore, can expect future benefits in revenues or profitability, cost avoidance, and high market growth. Thus, we hypothesize the following:

H9: ARMC is positively associated with firm performance.

3.2.7. ARMC, process innovation, and product innovation

In organizational settings, innovation is commonly defined as creating any new product, service, or process for an organization (Tushman & Nadler, 1986). A vast majority of research supports that firm innovation is a multidimensional phenomenon, and one way to conceptualize innovation as an organizational outcome is to separate product innovation from process innovation. Product innovation is “the novelty and meaningfulness of new products introduced to the market [in] a timely fashion” (Wang & Ahmed, 2004, p. 304). Product innovation is most often referred to as perceived newness, novelty, or uniqueness of products (Wang & Ahmed, 2004). It is also defined as any emerging technology or combination of emerging technologies that are exploited to produce goods or services (Bhoovaraghavan et al., 1996).

Process innovation is “the introduction of new production methods, management approach, and technology that can be used to improve production and management processes” (Wang & Ahmed, 2004, p. 305). Similarly, Trantopoulos et al. (2017) characterized process innovation as a novel and efficiency-improving endeavor to reduce the production costs of goods or services. In some studies, process innovation is considered a sub-element of technological innovation (Wang & Ahmed, 2004). While product innovation is centered on the market and primarily driven by customer needs, process innovation is internally oriented and primarily motivated by efficiency (Damanpour & Gopalakrishnan, 2001). Regardless of the differences between product and process innovation, prior research has suggested that they are closely related (Oke et al., 2007).

Recently, innovation has emerged as a key outcome of organizational agility, as evidenced by studies conducted by Battleson et al. (2016), Hoonsopon & Puriwat (2019), and Puriwat and Hoonsopon (2021). Considering this development, we contend that examining the impact of ARMC on innovation becomes particularly relevant when AI-powered capabilities are involved. AI can assist the innovation process by improving conditions for innovation and, in turn, facilitating the idea-generating process and overcoming information-processing constraints (Haefner et al., 2021). For example, machine learning-based methods are used to predict the most promising materials to test, providing firms with the ability to adapt. The need to change and adapt triggers creativity and the ability to innovate (Georgsdottir & Getz, 2004). Thus, we argue that AI indirectly influences firms’ innovation through ARMC.

The relationship between ARMC and innovation can be explained from several perspectives. First, from the creativity perspective, the flexibility of the existing knowledge structure has proven to be a source of new ideas (Georgsdottir & Getz, 2004). Mumford and Gustafson (1988) also argued that the flexible use of existing knowledge plays a critical role in generating new ideas. Since innovation overlaps with

Table 2

Summary of the interaction effects of environmental conditions.

	High AI-Enabled Automation	High AI-Enabled Analytics	High AI-Enabled Relational
High Env. Hostility	Low ARMC AI-enabled automation is desirable in a stable environment (H7a)	High ARMC AI-enabled analytics is desirable in a highly hostile environment (H7b)	Low ARMC AI-enabled relational capability is desirable in a stable environment (H7c)
High Env. Dynamism	Low ARMC AI-enabled automation is desirable in a stable environment (H8a)	High ARMC AI-enabled analytics is desirable in a highly dynamic environment (H8b)	High ARMC AI-enabled relational capability is desirable in a highly dynamic environment (H8c)

creativity, we infer that ARMC is related to process innovation. Second, prior organizational research has shown that when organizations are rigid and thus agility is low, company managers prefer conservative decisions, avoid risky behaviors, and consequently inhibit the process of innovation (Hisrich, 1990). Further, in a meta-analysis of organizational innovation, Damanpour (1991) found that among other factors, a positive managerial attitude toward change and the existence of special units that focus on change and adaptation processes facilitate innovation. In the context of technology capability, Zhou and Wu (2010) noted that “as a firm accumulates more technological capability, its absorptive capacity should increase, which leads to more product innovations” (p. 549). Firms with high ARMC are better at reacting to changes in market demand (Tallon & Pinsonneault, 2011). They can transform resources and capabilities to stimulate the development of new products (Hoon-sopon & Puriwat, 2019). Thus, we hypothesize the following:

H10: ARMC is positively associated with (a) process innovation, and (b) product innovation.

4. Research methodology

This study uses a two-stage survey to collect data from IT executives and business decision-makers (e.g., C-level executives, presidents, senior vice presidents, senior directors, directors, and managers or senior managers). These key informants are likely to be most informed about their firms' business and AI initiatives. We measured AI constructs, environmental factors, ARMC, and all control variables at one point in time (t). Then we measured firm performance, process innovation, and product innovation at the second point in time (t + 1). The first survey was conducted in November 2020, and the second was completed by the same respondents four months later. We chose the four-month gap to minimize response bias and ensure the accuracy of the information collected about the firms' performance. Conducting the second survey early in 2021 allowed us to capture up-to-date data and insights about firms' performance in the previous year.

We used a longitudinal study design primarily for three reasons: (1) to achieve correspondence with the temporal ordering of constructs in our research model; (2) to enhance our arguments for causality regarding the relationship between agility and organizational outcomes; and (3) to reduce the threats of common method bias. Data were collected through an online channel with organizations as our unit of analysis. Next, we discussed our methodological choices, encompassing concerns regarding measurement selection, sampling, response rate, and evaluation of potential non-response bias, following the recommendations of Hulland et al. (2018).

4.1. Measures

To the extent possible, we operationalized constructs by adapting existing scales to the context of our current studies. We captured responses to all items using multi-item reflective measures on a seven-point Likert scale. We then validated the measurement scales in a series of procedures to ensure content validity, construct validity, and reliability (Straub, 1989), as described later in this section. The details of measurement items can be found in Appendix A.

AI-Powered Capabilities: We measured three AI capabilities: AI-

enabled automation, AI-enabled analytics, and AI-enabled relational capabilities. Consistent with the literature (e.g., Huang et al., 2019; Huang & Rust, 2021a), we modeled each dimension as a first-order construct. Then we developed a new scale to operationalize AI capabilities based on insights from Cheatham et al. (2020) and used three items to measure AI-enabled automation. All three items measured the extent to which AI tools and applications are used for automation. For example, AI tools and applications are used to automate time-consuming activities, including claim processing, basic customer-service interaction, and inventory tracking. We measured AI-enabled analytics capability using nine items, including the use of AI to predict equipment failure, respond quickly to changing inventory levels, test nearly all potential scenarios before making a decision, predict the next-best product or service a customer is likely to buy, and simulate product or service delivery. To measure AI-enabled relational capability, we used four items that reflect the extent to which firms used AI to engage with their customers. These items were adapted from Lee et al. (2020).

To validate all items used in measuring AI-powered capabilities, we asked for feedback on our initial measurement items from two IS faculty members with experience in survey research methods. The feedback led us to review and modify several items. Next, the initial set of items and the predefined categories were submitted for a card-sorting test (Moore & Benbasat, 1991). Results from this test indicated that wording for some items needed additional clarity. After we conducted two rounds of card-sorting tests, the test resulted in a satisfactory classification of items into predefined categories.

ARMC: We measured ARMC using seven items that reflected how easily and quickly firms can address the opportunities in two areas: customer responsiveness and operational adjustment. These items were adapted from Tallon and Pinsonneault (2011).

Environmental Hostility and Dynamism: We measured environmental hostility using five items developed by Chen et al. (2014). The indicators assessed the existence of unfavorable external forces in firms' business environments (Zahra & Garvis, 2000). Environmental dynamism assessed the degree of rapid technological change, shifts in customers' preferences, and changes in competitor actions. The scale was adapted from Lee et al. (2015).

Firm Performance: We used self-reported items to measure firm performance. This approach is consistent with prior IT capability research investigating relative performance measures (e.g., Chen et al., 2014; Queiroz et al., 2018; Ravichandran et al., 2005). We adapted five items from Queiroz et al. (2018) to measure market share, revenues, sales growth, and profitability relative to competitors.

Process and Product Innovation: We modeled firms' product and process innovation as reflective, first-order constructs (Damanpour & Gopalakrishnan, 2001). Product innovation captured firms' capacity to innovate compared to all other competitors within the same industry concerning the level of novelty, utilization of cutting-edge technology, pace of product development, and number of new products or services introduced. On the other hand, process innovation denotes firms' technological competitiveness, speed in adopting the latest technology, and novelty in technology utilization. These items were adapted from Prajogo and Ahmed (2006).

Control Variables: We controlled for the effect of organizational factors (i.e., firm location, industry sector, firm size, and firm age) on

firm performance and innovation following prior work on the business value of IT (see [Chen et al., 2021](#); [Kulkarni et al., 2017](#)). *Firm's location* was a binary variable with 0 for UK companies and 1 for French companies. *Industry sector* was a binary variable with 0 for non-manufacturing firms and 1 for manufacturing firms. *Firm size* was measured as a firm-wide number of employees, and *firm age* was measured as the number of years a firm has been in business.

4.2. Survey development and administration

We pilot tested the final questionnaire with [Prolific.co](#) using the UK sample. Two screening criteria were used for the pilot study: (1) Subjects had supervisory responsibilities at their companies, and (2) their firms had adopted AI applications in recent years. After receiving 94 valid and complete responses in our pilot study, we reviewed and refined the items based on the results from the pilot study.

For the actual data collection, we asked a European professional market research company to contact IT and business decision-makers in diverse organizations. Our population of interest included firms that have adopted AI in Europe (i.e., France and the United Kingdom). For data collection from France, survey items were prepared in English and then translated into French. Despite the cultural differences, France and the UK share several commonalities, notably that both countries have stable economies and strong jurisdictions in Europe. They are also characterized by the same values, such as high autonomy and individual financial security, with a strong emphasis on achievement and individual initiative ([Mellahi, 2001](#)). According to OECD statistics ([Organization for Economic Co-operation & Development, 2019](#)), the rates of technology adoption in France and the UK were 90.17 and 95.85, respectively. Therefore, both countries are comparable in terms of technology adoption and business practices.

The samples for both countries were drawn from the panel members of the market research firm. We utilized the prescreening filters provided by this company to select appropriate respondents, such as IT or business executives. Additionally, we included two filter questions in our survey, measured on a five-point Likert scale, to ascertain whether respondents had sufficient knowledge about their companies' business and IT strategies. Only those who indicated adequate knowledge (above the cut-off point of three out of five) were eligible to participate in the study. The average IT knowledge of respondents was 3.88 (SD = 0.87), while their average business knowledge was 3.87 (SD = 0.86). These figures demonstrate a relatively balanced and comparable level of expertise in both areas. Four months after completing the first survey, the same respondents were contacted to complete the second survey. Of 225 respondents who completed the first survey, a total of 107 respondents completed the second survey for an effective response rate of 47.5 percent. We used two mechanisms to ensure that the same individual responded to both surveys for the same organization. First, each panelist had a unique identification number, allowing us to match the two data sets. Second, we asked several personal questions and matched their responses in both surveys. The matching technique provides support that the same individual responded to both surveys.

4.3. Sample characteristics

Data were collected from firms in France (68.2 %) and the U.K. (31.8 %). We provided a summary description of the sample in [Table 3](#). A comparison of earlier with later respondents from each country group regarding the firm's age and size indicated no differences, reducing concerns of non-response bias ([Armstrong & Overton, 1977](#)). Similar comparisons were made among participants from the two countries. The analysis indicated that the two countries' respondents were statistically similar regarding key study variables and firms' characteristics. The only variable that showed a slightly significant difference between the two countries was environmental hostility, with the UK sample reporting a slightly higher score of environmental hostility than the France

Table 3
Demographic respondents.

Demographic Information	Criteria	Total (%)
Firm's Geographic Location	France	73 (68.2)
	United Kingdom	34 (31.8)
Industry Type	Banking	8 (7.5)
	Computer/software	32 (29.9)
	Consulting	10 (9.3)
	Insurance	5 (4.7)
	Manufacturing	14 (13.1)
	Medicine/health	7 (6.5)
	Publishing/communications	8 (7.5)
	Hotel/restaurants	5 (4.7)
	Transportation	6 (5.6)
	Energy	4 (3.7)
	Other (constructions, trading and retail, commerce, education)	8 (7.5)
Company Age	< 5 years	2 (1.9)
	5–10 years	12 (12.1)
	6–15 years	22 (20.6)
	> 15 years	70 (65.4)
Number of Employees	<50	15 (14)
	50–200	15 (14)
	201–500	16 (15)
	501–1000	17 (15.9)
	1001–2500	15 (14)
	2501–10,000	14 (13.1)
Respondent Position	>10,000	14 (14)
	C-Suite Level Executive	17 (15.9)
	Vice President	2 (1.9)
	General Manager	16 (15)
	Business Unit Director	11 (10.3)
	IT Director	23 (21.5)
	Business Manager	20 (18.7)
	IT Manager or Project Manager	18 (16.8)

sample (t-value = -1.98; $p = .05$).

5. Data analysis and results

5.1. Measurement validation

We used Partial Least Squares (PLS) in the package WarpPLS 7.0 ([Kock, 2022](#)) to perform the data analysis. PLS is appropriate for the study because it handles second-order constructs better than covariance-based structural equation modeling (CB-SEM) does. It also assesses a measurement model for the theoretical model, makes no assumptions regarding data normality, and can handle small samples ([Kathuria et al., 2018](#)). The first stage of data analysis focused on the measurement properties of constructs. We assessed each construct's convergent and discriminant validity by factor-analyzing items grouped under their respective construct and the item-total correlations for all constructs. Several items were dropped due to a wording issue and/or low loadings. After we dropped this item, all remaining items loaded highly on the pre-specified construct (see Appendix B).

[Table 4](#) shows the descriptive statistics of all the constructs, including means, standard deviations, and latent variable coefficients. The average variance extracted (AVE) values for all reflective constructs

Table 4
Descriptive statistics and latent variable coefficients.

Construct	Mean	SD	CR	AVE
AI-enabled automation capability	5.28	1.14	0.88	0.72
AI-enabled analytics capability	4.94	1.34	0.96	0.74
AI-enabled relation capability	5.28	1.15	0.93	0.76
Adaptive response to market change (ARMC)	5.35	0.88	0.92	0.54
Environmental hostility	4.34	1.61	0.95	0.80
Environmental dynamism	5.31	1.14	0.87	0.78
Process innovation	5.03	1.07	0.94	0.83
Product innovation	4.96	1.07	0.93	0.75
Firm performance	4.99	1.19	0.95	0.79
Respondent IT knowledge	3.88	0.87	1.00	1.00
Respondent business knowledge	3.87	0.86	1.00	1.00

CR = Composite Reliability; AVE = Average Variances Extracted; Firm IT knowledge and business knowledge were measured using a 5-point Likert Scale.

were greater than the minimum recommended value of 0.50, indicating that the items satisfied the convergent validity. Further, to confirm the scale reliability and internal consistency of the constructs, we calculated the composite reliability scores (Fornell & Larcker, 1981). A composite reliability value of 0.70 or greater is considered acceptable. As reported in Table 4, the composite reliability scores for all the constructs were greater than 0.88, demonstrating that all constructs had adequate reliability assessment scores. To ensure the discriminant validity of constructs in the research model, we compared the square root of the AVE for each construct with the other correlation scores in the correlation matrix (see Table 5). As shown in Table 5, all AVE values' square roots were greater than the correlations between the respective constructs and other latent constructs (Fornell & Larcker, 1981).

5.2. Test of common method bias

In addition to using a longitudinal study design, we performed a series of statistical tests to alleviate common method concerns. First, we applied Harman's single-factor test using exploratory factor analysis (Podsakoff et al., 2003). The resulting unrotated component matrix showed that the measurement items did not load on a single factor. A single factor accounted for only 38.02 % of the total variance. Second, we examined common method variance (CMV) using a marker variable technique (Lindell & Whitney, 2001) for the variables collected at one point in time (t). In this technique, a marker variable is a scale that is theoretically unrelated to at least one other scale in the questionnaire. The marker variable's mean correlation with other variables in the study (r_M) is an estimate of the extent of CMV in the sample. We computed CMV-adjusted correlations between the variables of interest (r_A) by partialling out r_M from the uncorrected bivariate correlation (r_U) using the following formula $r_A = (r_U - r_M) / (1 - r_M)$ (Malhotra et al., 2006) and we estimated the difference between r_U and r_M using a t-statistic (Bhattacharjee & Hikmet, 2007). In our study, we used respondents' current state of mind (i.e., "my mind feels unfocused right now") as the marker variable. This variable had a mean absolute correlation of 0.07 with the items used in this study. The mean change in correlation due to the CMV-adjusted correlations was very small (0.03). These results suggest the lack of CMV bias in our sample.

5.3. Structural model

We used WarpPLS 7.0 to assess the structural model. Since ARMC is conceptualized as a second-order aggregate construct, we generated latent variable scores for each of its first-order dimensions, which were then used as reflective measures of the second-order aggregate constructs (Polites & Karahanna, 2012). Fig. 2 displays the final result of the structural model, and Table 6 presents the results of hierarchical regression analyses. Model 1a specifies the direct effects of AI-powered capabilities on the dependent variables. Model 1b tests the interaction effects model. Models 2a, 3a, and 4a specify the effects of control

variables on the dependent variables, and Models 2b, 3b, and 4b explore the effects of ARMC on the dependent variables.

As illustrated in Fig. 2, AI-enabled automation was positively associated with AI-enabled analytics and AI-enabled relational capabilities, and AI-enabled analytics capability was positively associated with AI-enabled relational capability. Thus, H1, H2, and H3 were supported. In Model 1a, the direct effects of AI-enabled automation, AI-enabled analytics capability, and AI-enabled relational capability on ARMC were significant. However, adding the interaction terms to the model significantly altered the direct effects of AI-powered capabilities.³ Hence, we used Model 1b as our final model. The effect of AI-enabled automation and AI-enabled analytics on ARMC remained significant even after the interaction terms were added to the model, supporting H4 and H5. However, the effect of AI-enabled relational capability became insignificant, failing to support H6.

As expected, environmental hostility negatively moderated the relationship between AI-enabled automation and ARMC, supporting H7a. Environmental hostility positively moderated the relationship between AI-enabled analytics capability and organizational agility, supporting H7b. As we hypothesized, the interaction between AI-enabled relational capability and environmental hostility was negative and significant. Thus, H7c was supported. However, the interactions between AI-enabled automation and environmental dynamism, and between AI-enabled analytics and environmental dynamism, were not significant. Thus, H8a and H8b were not supported. The interaction between AI-enabled relational and environmental dynamism was significant, supporting H8c. When the interaction effects of environmental conditions were included, the adjusted R^2 of ARMC was significantly increased by seven percent ($p < .001$), indicating a medium-to-large-effect size ($f^2 = 0.22$).⁴ For better visual clarity, these interaction effects are presented as a 3D graph in Appendix C (Fig. C1).

Further, ARMC was positively associated with firm performance, supporting H9. Consistent with H10, ARMC had a significant effect on both process and product innovation. To further validate the indirect effects of ARMC, we tested the significance of both indirect and moderated indirect effects (if the relationship consists of both mediation and moderator). As shown in Table 7, 10 out of 18 conditional indirect effects of AI-powered capabilities on organizational outcomes were significant, suggesting meaningful moderating roles of environmental conditions.

5.4. Test of endogeneity

When causal effects are examined, endogeneity issues may arise from omitted variable bias. To address the potential endogeneity issues, we adopted an instrumental variable approach introduced by Kock (2022) in WarpPLS. A valid instrumental variable should be uncorrelated with the error (i.e., satisfy the exclusion restriction) and correlated with the endogenous regressor (i.e., satisfy the relevance condition) (Tan et al., 2017). In other words, the instrument should affect the dependent variables (i.e., firm performance and innovation) only through the endogenous regressor (ARMC). We proposed IT infrastructure compatibility as a valid instrumental variable (i.e., it highly correlates with ARMC ($r = 0.58, p < .001$) but does not correlate with the organizational

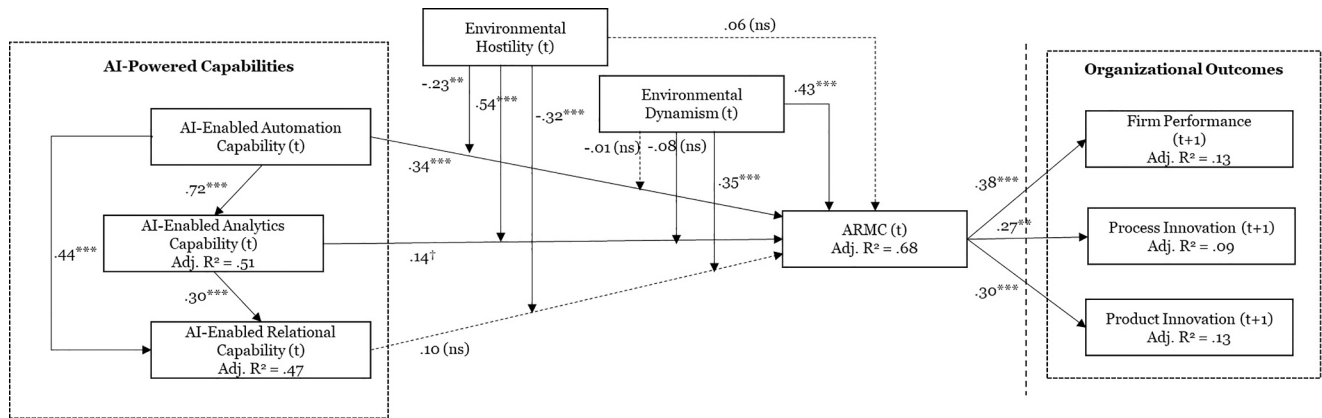
³ The presence of the interaction term alters the strength or direction of the relationships between the independent variables and the dependent variable, rendering the main effects non-significant. This result suggests that the relationships between the independent variables and dependent variables vary across different levels of moderators. Thus, the relationships should be interpreted as "conditional" effects at the different values of the included moderator in the model, and the main effects become less relevant (Frazier et al., 2004).

⁴ Effect size (f^2) is calculated by the formula $(R^2_{\text{full}} - R^2_{\text{partial}}) / (1 - R^2_{\text{full}})$ (Mathieson et al. 2001). Cohen (1988) suggested 0.02, 0.15, and 0.35 as operational definitions of small-, medium-, and large-effect sizes, respectively.

Table 5
Correlation matrix of the focal constructs.

Construct	1	2	3	4	5	6	7	8	9	10	11	12	13
1 AI-enabled automation	0.85												
2 AI-enabled analytics	0.71***	0.86											
3 AI-enabled relation	0.66***	0.62***	0.87										
4 ARMC	0.70***	0.61***	0.62***	0.92									
5 Env. Hostility	0.31**	0.46***	0.20*	0.26**	0.90								
6 Env. Dynamism	0.50***	0.67***	0.55***	0.67***	0.21*	0.88							
7 Firm performance	0.24*	0.40***	0.31**	0.40***	0.20*	0.27**	0.89						
8 Process innovation	0.31**	0.35***	0.22*	0.31**	0.24*	0.19	0.58***	0.91					
9 Product innovation	0.32***	0.35***	0.31**	0.35***	0.24*	0.25**	0.64***	0.77***	0.87				
10 Industry type	-0.22*	-0.12	-0.25*	-0.22*	0.04	-0.11	-0.15	-0.22*	-0.24*	–			
11 Firm age	-0.05	-0.19*	-0.01	-0.06	-0.27**	-0.08	-0.02	-0.12	-0.09	0.00	–		
12 Firm size	-0.04	-0.06	0.11	0.07	-0.15	-0.01	0.11	0.01	0.02	-0.23*	0.26**	–	
13 Firm location	0.01	-0.14	0.05	-0.01	-0.19	-0.06	-0.05	-0.03	-0.04	-0.16	0.36***	0.17	–

Notes: Square roots of average variances extracted (AVEs) are shown on the diagonal; ***p < .001; **p < .01; *p < .05.



***p < .001; **p < .01; *p < .05; †p < .10; Control variables are suppressed for illustrative purposes.

Two dimensions of ARMC: customer responsiveness and operational adaptiveness, were positively correlated ($r = .69$; $p < .001$). Their reflective construct loadings were .92 and .92, respectively.

Fig. 2. Structural model.

Table 6
Statistical results (standardized path estimates).

	DV: ARMC		DV: Firm Performance		DV: Process Innovation		DV: Product Innovation	
	Model 1a	Model 1b	Model 2a	Model 2b	Model 3a	Model 3b	Model 4a	Model 4b
Adj. R ²	0.53	0.68	0.00	0.13	0.03	0.09	0.04	0.13
Direct Effects								
AI-enabled automation	0.42***	0.34***						
AI-enabled analytics capability	0.16*	0.14†						
AI-enabled relational capability	0.24**	0.10 (ns)						
ARMC				0.38***		0.27**		0.30***
Env. Hostility		0.06 (ns)						
Env. Dynamism		0.43***						
Moderating Effects								
AI-enabled auto x Env. Hostility		-0.23**						
AI-enabled analytics x Env. Hostility		0.54***						
AI-enabled relational x Env. Hostility		-0.32***						
AI-enabled auto x Env. Dynamism		-0.01 (ns)						
AI-enabled analytics x Env. Dynamism		-0.08 (ns)						
AI-enabled relational x Env. Dynamism		0.35**						
Control Variables								
Firm location			0.08	-0.07 (ns)	-0.03 (ns)	-0.02 (ns)	-0.04 (ns)	-0.05 (ns)
Industry type			-0.14†	-0.06 (ns)	-0.23**	-0.17*	-0.26**	-0.19*
Firm size			0.09	0.08 (ns)	-0.02 (ns)	-0.02 (ns)	-0.01 (ns)	-0.02 (ns)
Firm age			-0.02 (ns)	0.01 (ns)	-0.10 (ns)	-0.09 (ns)	-0.09 (ns)	-0.05 (ns)

***p < .001; **p < .01; *p < .05; †p < .10; The adjusted R² of the model with the direct effects of AI-powered capabilities and environmental conditions was 0.61 (for brevity, this model is not included in the table).

outcomes). The results show that the instrumental variable was insignificant, suggesting that endogeneity is not a major issue in our study.

Further, we conducted the Durbin-Wu-Hausman test in Stata to

assess whether endogeneity was an issue. First, we regressed ARMC on all exogenous and control variables, including IT infrastructure compatibility. Then, we used the residual of the regression as an

Table 7Analysis of indirect and conditional indirect effects.⁵

Relationship	(Conditional) Indirect Effects (p-value)	Type of Mediation/ Indirect Effects
AI-enabled automation → AI-enabled analytics → AI-enabled relational	0.22***	Partial mediation
AI-enabled automation → AI-enabled analytics → ARMC	0.10*	Partial mediation
AI-enabled automation → AI-enabled relational → ARMC	0.04	No mediation
AI-enabled analytics → AI-enabled relational → ARMC	0.03	No mediation
AI-enabled automation → ARMC → firm performance	0.14*	Indirect only, full mediation
AI-enabled analytics → ARMC → firm performance	0.05	No mediation
AI-enabled relational → ARMC → firm performance	0.04	No mediation
AI-enabled automation*environmental hostility → ARMC → firm performance	-0.09 [†]	Indirect only, moderated full mediation
AI-enabled analytics*environmental hostility → ARMC → firm performance	0.21***	Moderated partial mediation
AI-enabled relational*environmental hostility → ARMC → firm performance	-0.12*	Indirect only, moderated full mediation
Environmental hostility → ARMC → firm performance	0.03	No mediation
AI-enabled automation*environmental dynamism → ARMC → firm performance	0.01	No moderated mediation
AI-enabled analytics*environmental dynamism → ARMC → firm performance	-0.05	No moderated mediation
AI-enabled relational*environmental dynamism → ARMC → firm performance	0.13*	Indirect only, moderated full mediation
Environmental dynamism → ARMC → firm performance	0.16**	Partial mediation
AI-enabled automation → ARMC → process innovation	0.10 [†]	Indirect only, full mediation
AI-enabled analytics → ARMC → process innovation	0.04	No mediation
AI-enabled relational → ARMC → process innovation	0.03	No mediation
AI-enabled automation*environmental hostility → ARMC → process innovation	-0.06	No moderated mediation
AI-enabled analytics*environmental hostility → ARMC → process innovation	0.14*	Moderated partial mediation
AI-enabled relational*environmental hostility → ARMC → process innovation	-0.09 [†]	Indirect only, moderated full mediation
Environmental hostility → ARMC → process innovation	0.02	No mediation
AI-enabled automation*environmental dynamism → ARMC → process innovation	0.00	No moderated mediation
AI-enabled analytics*environmental dynamism → ARMC → process innovation	-0.04	No moderated mediation
AI-enabled relational*environmental dynamism → ARMC → process innovation	0.09 [†]	Indirect only, moderated full mediation
Environmental dynamism → ARMC → process innovation	0.11*	Partial mediation
AI-enabled automation → ARMC → product innovation	0.11*	Indirect only, full mediation
AI-enabled analytics → ARMC → product innovation	0.04	No mediation
AI-enabled relational → ARMC → product innovation	0.03	No mediation

Table 7 (continued)

Relationship	(Conditional) Indirect Effects (p-value)	Type of Mediation/ Indirect Effects
AI-enabled automation*environmental hostility → ARMC → product innovation	-0.07	No moderated mediation
AI-enabled analytics*environmental hostility → ARMC → product innovation	0.17**	Moderated partial mediation
AI-enabled relational*environmental hostility → ARMC → product innovation	-0.10 [†]	Indirect only, moderated full mediation
Environmental hostility → ARMC → product innovation	0.02	No mediation
AI-enabled automation*environmental dynamism → ARMC → product innovation	0.00	No moderated mediation
AI-enabled analytics*environmental dynamism → ARMC → product innovation	-0.04	No moderated mediation
AI-enabled relational*environmental dynamism → ARMC → product innovation	0.11*	Moderated partial mediation
Environmental dynamism → ARMC → product innovation	0.13*	Partial mediation

Note: Only indirect effects for paths with two segments were reported; ***p < .001; **p < .01; *p < .05; [†]p < .10.

⁵ We also measured the direct effects of AI-enabled automation capability, AI-enabled analytics capability, and AI-enabled relational capability on firm performance, process innovation, and product innovation. The only significant capability was AI-enabled analytics capability to all three dependent variables, with the path coefficients as follows: AI-enabled analytics to firm performance = 0.44, $p < .001$; AI-enabled analytics to process innovation = 0.29; $p < .001$; and AI-enabled analytics to product innovation = 0.28; $p < .001$. Adding ARMC and interaction terms to the model reduced these direct effects to 0.27, 0.27, and 0.26, respectively (all are significant at $p < .01$).

additional regressor in our hypothesized equations. The parameter estimate for the residual was not significant, indicating that ARMC was not endogenous in our study. Further, the Durbin (score) ($X^2 = 0.78$; $p = .38$) and the Wu-Hausman test were not significant ($p = .40$). These insignificant results indicate that ARMC was exogenous.

6. Discussion

The primary objective of this study is to explore how AI-powered capabilities can add value to organizations by enhancing their ARMC. We have proposed that three AI-powered capabilities—AI-enabled automation, AI-enabled analytics, and AI-enabled relational capabilities—play pivotal roles in driving ARMC. Drawing on the perspectives of organizational agility and the dynamic capability framework, we conceptualize ARMC as firms' capacity to swiftly recognize and adapt to market shifts, encompassing customer responsiveness and operational adjustment as their core competencies. We also examine the moderating effects of environmental conditions (environmental hostility and dynamism) on the relationships between AI-powered capabilities and ARMC.

Our findings reveal that the relationships between AI-powered capabilities and ARMC depend on environmental conditions. Specifically, the findings show that environmental hostility negatively moderates the relationship between AI-enabled automation and ARMC, as well as between AI-enabled relational capability and ARMC. However, it positively moderates the relationship between AI-enabled analytics and ARMC. Additionally, the findings demonstrate that environmental dynamism positively moderates the relationship between AI-enabled relational capability and ARMC. Finally, our study establishes significant positive relationships between ARMC and both firm performance and innovation. These findings have important implications for both research and practice.

6.1. Implications for research

Our study makes several contributions to research on the business value of IT. First, the results provide robust empirical evidence supporting the idea that AI can enhance organizational responsiveness and adaptability in the face of market changes. The hype around AI is causing many companies to prematurely adopt AI applications without fully grasping the true value of AI (Brock & von Wangenheim, 2019; Davenport, 2018). However, directing AI toward the key business competencies can strengthen firms' ability to be adaptive, agile, and responsive to market changes, thereby positioning firms for sustainable growth and success in dynamic business environments.

Second, our findings suggest a positive association between AI-enabled automation and ARMC. By leveraging AI-enabled automation tools, such as RPA, firms can automate repetitive and time-consuming business processes, resulting in improved productivity, faster market entry for products and services, and enhanced responsiveness to market opportunities. However, our findings indicate that AI-enabled automation is more suitable for stable environments with well-defined processes and relatively predictable changes, where firms can produce and sell homogenous products in large volumes (Lee et al., 2015). Thus, AI-enabled automation efficiently streamlines production and operational processes, ensuring consistency and cost-effectiveness while minimizing human errors and improving overall efficiency.

Third, our findings also reveal that the effect of AI-enabled analytics on ARMC is significantly altered once the moderating effect of environmental hostility is added to the model. Specifically, in highly hostile environments where uncertainties and disruptions are prevalent, the impact of AI-enabled analytics on ARMC becomes even more pronounced. AI-enabled analytics can assist organizations in processing vast amounts of complex data and extracting actionable insights, allowing them to gain a deeper understanding of market dynamics and customer behavior (Huang et al., 2019). In such hostile conditions, the ability to quickly identify emerging trends and adapt business strategies accordingly becomes a critical advantage for firms seeking to control the inherent risk. Insights enabled by AI overcome the challenges of obtaining information in a hostile environment, enabling evidence-based decisions in complex situations (Fernández-Rovira et al., 2021). Thus, AI-enabled analytics capability allows firms to develop planned strategies in response to constant threats posed by the environment.

Fourth, our results demonstrate that environmental hostility negatively moderates the relationship between AI-enabled relational capability and ARMC. AI tools with relational capability utilize existing conversation data to understand a wide range of customers and employ machine learning and NLP to grasp the context, continuously enhancing AI's responses and interactions with customers (Davenport & Ronanki, 2018). These diverse digital options enabled by AI relational capability can potentially drive ARMC both at market and operational levels as organizations become more effective in responding to customers, leading to changes in internal business processes. However, many AI systems designed for this purpose are still experimental (Huang & Rust, 2020). Thus, in highly hostile environments where the risk is too high and can be costly (Mitchell et al., 2011), the effectiveness of AI-enabled relational capability may be hampered, and organizations should carefully evaluate its implementation and consider alternative strategies to enhance ARMC under such conditions.

Fifth, we found that environmental dynamism does not moderate the relationship between AI-enabled automation and ARMC, as well as between AI-enabled analytics and ARMC. However, we observed a positive and significant moderation effect of environmental dynamism on the relationship between AI-enabled relational capability and ARMC. In environments characterized by high dynamism, firms often face significant unpredictability regarding their customers' tastes and preferences (Lee et al., 2015). Consequently, they need to continuously engage in competitive actions to attract and retain customers. In this context, AI-enabled analytics can aid in establishing a relational capability,

which, in turn, facilitates the development of ARMC in a highly dynamic environment. By leveraging AI-enabled relational capability, organizations can adapt their customer interactions and responses in real time, enabling them to stay responsive to the ever-changing market conditions and better meet customer needs and preferences.

Lastly, our findings strongly support a positive relationship between ARMC and firm performance, process innovation, and product innovation. From the dynamic capabilities perspective, ARMC empowers firms to seize new opportunities (Teece et al., 2016). However, being highly responsive and adaptive to the environment can come with costs. Consequently, the necessity for ARMC, and thus organizational agility, arises when substantial uncertainty and dynamism are present in the business environment (Chakravarty et al., 2013; Teece et al., 2016). Moreover, AI can potentially revolutionize the way firms operate, transforming an entire organization's rigid mindset into a more responsive and adaptive approach. This transformation provides an opportunity to explore problems from multiple perspectives, paving the way for innovative and novel solutions.

6.2. Practical and societal implications

Our findings provide several practical implications. First, executives exploring the potential of AI can use our results to guide their AI investments. Our study shows that different functions of AI can enhance ARMC, depending on the environmental conditions. With the uncertainty surrounding AI, executives would be wise to systematically evaluate their business needs to determine which areas could benefit from AI. The three AI-powered capabilities discussed in this study may help organizations to broaden their digital options and flexibly respond to market opportunities.

Our findings suggest that not all AI capabilities hold the same significance. AI-enabled automation can aid organizations in cost reduction and standardizing routine tasks, particularly in stable environments. On the other hand, AI-enabled analytics capability may enhance firms' ARMC by processing, evaluating, and analyzing information, proving more valuable in a hostile environment. Lastly, AI-enabled relational capability is key in fostering ARMC by actively engaging with customers and continuously learning from these interactions, making the process instrumental for adaptability in highly dynamic environments. By leveraging the diverse capabilities of AI, organizations can effectively adapt to changing environments and improve their performance.

Our findings also suggest that nurturing their capacity for swift and effective responses to market shift enables organizations to seize new opportunities. This study highlights the positive impact of ARMC on innovation outcomes. Although implementing AI can be challenging, AI systems can drive ARMC, leading to both process and product innovation, as well as improved performance outcomes. Therefore, AI systems enable firms to explore novel approaches to problem-solving, providing them with the flexibility to respond to threats and opportunities (Haefner et al., 2021). With such flexibility, firms can readily identify emerging market demands and create new products/services to meet those needs.

Our study indicates that AI capabilities are transforming the dynamics between humans and machines, the methods of data analysis, and the decision-making process from a societal standpoint. Although AI initiatives can greatly influence customer experience and workforce skills (Butner et al., 2020), how these changes will unfold in the long run is currently unknown. For instance, past research has shown that under high levels of environmental hostility and dynamism, IT resources and competencies are necessary so that managers can actively sense opportunities for competitive actions (Lee et al., 2015). AI has transformed how organizations perceive and respond to such challenges since AI-powered capabilities provide a more flexible and adaptive approach. As AI technologies continue to advance, they are likely to play an even more significant role in shaping the future of business and society. However, alongside the tremendous potential are ethical, regulatory,

and privacy concerns that need to be carefully addressed to ensure responsible and sustainable AI deployment (Whittlestone et al., 2019). Thus, continued research on the impacts of AI and its potential challenges is essential to harnessing its full benefits while mitigating potential risks.

6.3. Limitations and future research directions

This research has some limitations that suggest directions for future work. First, as mentioned earlier, organizational agility is a comprehensive and multifaceted concept. While we derived the concept of ARMC from the broader agility framework, we encourage future research to explore the wider aspects of organizational agility. This expanded investigation could uncover additional dimensions and potential intersections that could enhance our understanding of how AI-powered capabilities contribute to firms' responsiveness and adaptability. Further, as Kalaighnanam et al. (2021) observed, sensemaking, speed, and iteration define agility. Our concept of ARMC does not explicitly contain iteration; however, it aligns with the crucial elements of sensemaking and speed, which are integral components of organizational agility. We call for future research to conduct a longitudinal study, explicitly observing all three aspects of agility in the study.

Second, while many researchers have defined agility as encompassing speed and flexibility (e.g., Kalaighnanam et al., 2021), Prange (2021) suggests that agility is not reliant solely on speed as a defining characteristic. Specifically, according to Prange, agility requires both the resilience derived from a reliable core and the dynamism anchored in ongoing change. Our conceptualization of ARMC does not explicitly elaborate on the concept of agility as slowness. Future research should explore different aspects of agility to lead to a better understanding of how agility, including ARMC, can contribute to organizational success in varying environments.

Third, we conceptualize each AI-powered capability as a first-order construct by following the existing concept of multiple AIs (e.g., Huang et al., 2019; Huang & Rust, 2021a). However, the relationships among AI-powered capabilities may be more complex than we have hypothesized. For instance, interaction effects could occur among AI capabilities. A larger sample size would be required to test such complex relationships. To examine these complexities further, exploratory research could achieve a much stronger theoretical foundation. Understanding the dependencies among a wider capability system in an organization could be an interesting and important agenda for future research in this area, especially under different environmental conditions.

Fourth, although we conducted a two-stage survey to address the threats of common method bias, we still used the same respondent for both our independent and dependent variables. Although common

method bias does not appear to threaten the validity of our results statistically, using the same respondent might have upwardly biased our results. Thus, we recommend that future studies consider research designs that allow data collection from multiple respondents within an organization. Future research could also complement the self-reported measures of performance with more objective performance data. Lastly, our data were collected from firms located in France and the U.K., and our sample includes a broad spectrum of industry groups. Although we controlled for the effect of location and industry types, investigating data from a more homogeneous population could result in different conclusions.

7. Conclusion

Our research investigates how AI-powered capabilities can add value to organizations through their capacity to directly affect how firms identify and respond to market changes. We offer a concept of ARMC derived from organizational agility and dynamic capabilities perspectives. We also investigate the moderating role of environmental conditions in influencing the relationship between AI-powered capabilities and ARMC. Our findings indicate that the effects of AI-enabled automation, AI-enabled analytics, and AI-enabled relational capability on ARMC are contingent on two environmental conditions—environmental hostility and environmental dynamism. Our research provides a theory-based understanding of AI's impact on organizations. In addition, we endeavor to establish a foundational framework for future theory-driven studies to gain a comprehensive understanding of how AI benefits organizations. This understanding can contribute to improved organizational performance and foster additional competitive actions.

CRedit authorship contribution statement

Yulia Sullivan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.
Samuel Fosso Wamba: Writing – original draft, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. . Measurement items

AI-Enabled Automatic Capability (New scale, measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).

1. In our firm, AI tools and applications are used to automate time-consuming activities, including claim processing, basic customer-service interaction, and inventory tracking.
2. In our firm, AI tools and applications are used to save staff time by allowing them to focus on higher-value activities.
3. In our firm, AI tools and applications are used to automate routine as well as back-office tasks.

AI-Enabled Analytics Capability (New scale, measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).
 In our organization, AI tools and applications allow us to...

1. ...predict more accurately potential equipment or service failures.
2. ...respond more quickly to changing stock or inventory levels.
3. ...test nearly all potential scenarios before making a decision.
4. ...rapidly analyze millions of combinations to help employees test the efficiency of different decisions under different conditions.

5. ...for our executive teams understand likely ranges of outcomes resulting from acquisitions of suppliers.
6. ...*accurately price our products or services (dropped)*
7. ...predict more accurately the next-best product or service a customer is likely to buy.
8. ...forecast our customer traffic (e.g., in our stores or on our websites).
9. ...for our middle managers simulate product or service delivery to identify high-value products or services.

AI-Enabled Relational Capability (New scale, adapted from [Lee et al., 2020](#), measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).

1. Our AI-based system offers various decision-making tools that enable us to manage our relationship with our customers.
2. Our AI-based system offers various tools, such as chatbots, that enable us to support our interactions with our customers.
3. Our AI-based system offers various tools that enable us to examine trends in the data for managing our interaction with our customers.
4. We use virtual agents (e.g., chatbots and machine learning) to respond to our customers' queries.

Adaptive Response to Market Changes (ARMC) (Source: [Tallon & Pinsonneault, 2011](#)).

How easily and quickly can your firm perform the following actions? (measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).

Customer responsiveness

1. Respond to changes in aggregate consumer demand.
2. Customize a product or service to suit an individual customer.
3. React to new product or service launches by competitors.
4. Introduce new pricing schedules in response to changes in competitors' or markets' prices.

Operation adjustment

1. Expand into new regional or international markets.
2. Change (i.e., expand or reduce) the variety of products/services available for sale.
3. Adopt new technologies to produce better, faster, and cheaper products and services.
4. *Switch suppliers to avail of lower costs, better quality, or improved delivery times (dropped)*

Environmental Hostility (Source: [Chen et al., 2014](#), measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).

1. The survival of our firm is currently threatened by the scarce supply of labor.
2. The survival of our firm is currently threatened by the scarce supply of materials.
3. The survival of our firm is currently threatened by tough price competition.
4. The survival of our firm is currently threatened by tough competition in product/service quality.
5. The survival of our firm is currently threatened by tough competition in product/service differentiation.

Environmental Dynamism (Source: [Lee et al., 2015](#), measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).

1. Technological changes in our industry are rapid.
2. Customers' product/service preferences change rapidly.
3. *The actions of competitors in our major markets are changing rapidly (dropped)*

Innovation (Source: [Prajogo & Ahmed, 2006](#)).

Indicate your firm's ability to innovate during the COVID-19 crisis relative to all other competitors in the same industry (1 = far below average to 7 = far above average) (note: the COVID-19 statement was used in the questionnaire because data were collected during the COVID pandemic).

Product innovation

1. Level of newness (novelty)
2. Use of the latest technology
3. *Speed of product development (dropped)*
4. Number of new products/services

Process innovation

1. Technology competitiveness
2. Speed of adopting the latest technology
3. Novelty of the technology use

Firm Performance (Source: [Queiroz et al., 2018](#), measured on a seven-point Likert scale ranging from 1 = strongly disagree to 7 = strongly agree).
To what extent do the following statements reflect your firm's current situation in the last 12 months?

1. We are more profitable than our competitors.
2. Our sales growth exceeds that of our competitors.
3. Our revenue growth exceeds that of our competitors.
4. Our market share growth exceeds that of our competitors.
5. Overall, our performance is better than our competitors.

Appendix B. . WarpPLS loadings and Cross-Loadings

	AI_RL	AI_ANL	AI_AUTO	ARMC	E_HOST	E_DYN	PROC_INN	PROD_INN	FP	IND	C_SIZE	C_SIZE	LOC
AI_RL1	0.87	−0.02	−0.07	0.02	0.17	0.11	−0.01	0.01	−0.03	−0.04	0.11	0.01	−0.07
AI_RL2	0.89	−0.03	0.13	−0.08	−0.04	0.09	0.02	−0.10	0.09	0.06	−0.05	0.03	0.03
AI_RL3	0.91	0.27	−0.23	0.09	−0.10	−0.05	0.05	−0.08	−0.02	−0.08	0.03	−0.13	0.04
AI_RL4	0.81	−0.25	0.20	−0.03	−0.04	−0.16	−0.05	0.19	−0.04	0.08	−0.10	0.10	−0.01
AI_ANL1	−0.24	0.88	0.14	−0.18	0.01	0.38	0.04	−0.05	0.04	0.04	−0.05	0.17	−0.07
AI_ANL2	0.16	0.86	0.20	−0.21	0.04	0.09	−0.03	0.07	0.02	0.08	0.04	0.01	−0.03
AI_ANL3	−0.22	0.87	0.19	−0.03	−0.03	−0.25	−0.16	0.18	−0.09	−0.05	0.02	−0.02	0.05
AI_ANL4	0.06	0.81	0.38	0.00	−0.06	−0.21	−0.27	0.26	−0.08	0.01	−0.09	0.03	0.05
AI_ANL5	0.21	0.84	−0.33	0.28	−0.01	0.03	0.00	0.07	−0.08	−0.04	0.18	−0.06	−0.14
AI_ANL7	0.02	0.90	−0.09	−0.03	0.06	0.08	0.11	−0.16	0.11	−0.04	−0.01	−0.09	−0.01
AI_ANL8	−0.11	0.84	−0.26	0.17	−0.08	−0.16	0.05	−0.09	0.01	−0.07	−0.11	−0.12	0.10
AI_ANL9	0.14	0.89	−0.22	0.01	0.06	0.01	0.24	−0.25	0.06	0.07	0.01	0.07	0.04
AI_AUTO1	0.15	−0.41	0.85	−0.05	−0.03	0.01	0.07	−0.08	0.05	0.16	−0.09	0.08	0.10
AI_AUTO2	0.07	−0.14	0.92	0.07	−0.03	−0.04	−0.12	0.10	−0.01	0.00	0.06	0.01	−0.06
AI_AUTO3	−0.25	0.63	0.76	−0.03	0.07	0.05	0.07	−0.03	−0.05	−0.17	0.02	−0.10	−0.04
ARMC_CR	−0.08	0.23	−0.21	0.92	−0.03	0.05	−0.12	0.12	0.00	−0.06	0.02	−0.06	−0.07
ARMC_OA	0.08	−0.23	0.21	0.92	0.03	−0.05	0.12	−0.12	0.00	0.06	−0.02	0.06	0.07

(continued on next page)

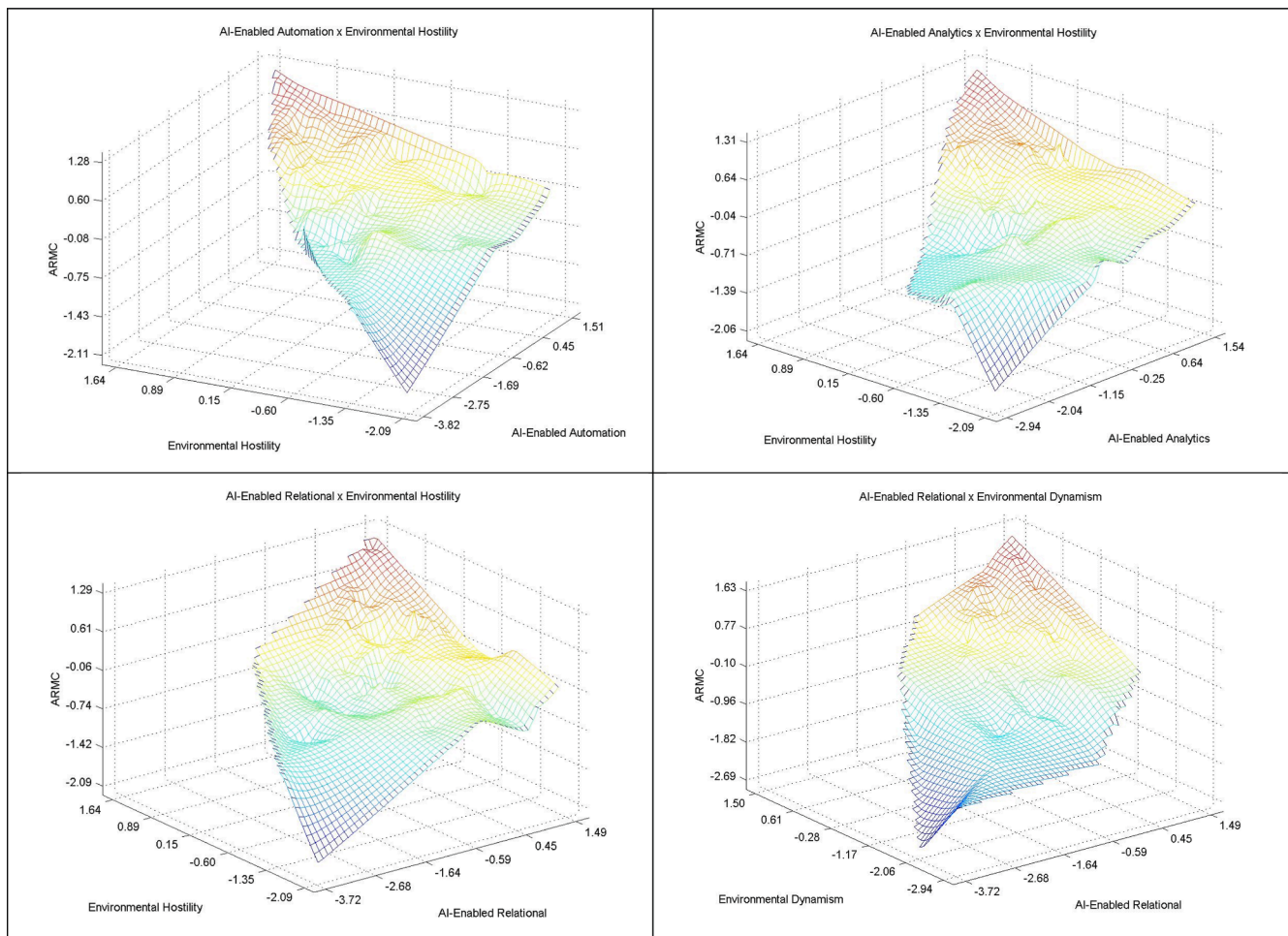


Fig. C1. 3D Plots for moderating effects.

(continued)

	AI_RL	AI_ANL	AI_AUTO	ARMC	E_HOST	E_DYN	PROC_INN	PROD_INN	FP	IND	C_AGE	C_SIZE	LOC
E_HOST1	-0.25	0.31	-0.11	0.02	0.88	0.03	-0.10	0.10	-0.04	-0.06	-0.11	0.08	0.09
E_HOST2	-0.06	0.24	-0.14	-0.14	0.88	0.13	-0.06	0.03	0.11	-0.09	-0.06	0.08	0.05
E_HOST3	0.17	-0.39	0.21	-0.10	0.90	0.13	0.05	-0.18	0.08	0.05	0.07	-0.09	-0.10
E_HOST4	0.03	0.05	-0.10	0.15	0.92	-0.18	-0.05	0.19	-0.16	-0.02	0.01	-0.03	0.03
E_HOST5	0.10	-0.20	0.13	0.07	0.90	-0.10	0.15	-0.14	0.01	0.11	0.10	-0.04	-0.07
E_DYN1	0.00	-0.34	0.09	-0.17	-0.05	0.88	0.03	0.05	-0.06	-0.01	-0.02	0.07	-0.05
E_DYN2	0.00	0.34	-0.09	0.17	0.05	0.88	-0.03	-0.05	0.06	0.01	0.02	-0.07	0.05
PROC_INN1	0.14	0.03	-0.19	0.07	-0.01	-0.02	0.93	0.01	-0.05	0.01	0.02	0.01	0.03
PROC_INN2	-0.10	-0.10	0.12	-0.04	0.00	0.05	0.90	-0.05	-0.04	-0.01	0.01	-0.02	-0.04
PROC_INN3	-0.04	0.07	0.08	-0.04	0.01	-0.04	0.90	0.04	0.09	-0.01	-0.03	0.01	0.01
PROD_INN1	0.14	0.08	-0.15	0.05	-0.01	-0.08	-0.23	0.87	0.23	-0.09	0.03	0.10	-0.09
PROD_INN2	0.10	-0.15	0.00	0.04	-0.07	0.03	0.36	0.86	-0.30	0.10	0.02	0.07	-0.08
PROD_INN4	-0.24	0.07	0.15	-0.08	0.08	0.05	-0.14	0.88	0.07	-0.01	-0.05	-0.17	0.18
FP1	-0.13	-0.06	0.18	0.01	-0.01	-0.08	0.16	-0.26	0.88	-0.10	-0.03	-0.02	0.02
FP2	0.00	0.03	-0.06	0.08	0.03	-0.07	0.22	-0.22	0.93	0.02	0.04	-0.04	-0.04
FP3	-0.11	-0.07	0.09	-0.09	0.15	0.11	-0.21	0.07	0.90	-0.07	0.05	-0.09	0.00
FP4	0.19	0.09	-0.17	-0.01	-0.11	0.02	-0.19	0.29	0.85	0.14	-0.08	0.15	0.04
FP5	0.06	0.01	-0.03	0.01	-0.07	0.01	0.01	0.14	0.90	0.02	0.02	0.00	-0.03

Appendix C. . 3D Plots for the moderating effects of environmental conditions

Fig. C1.

References

- Albrecht, T., Rausch, T. M., & Derra, N. D. (2021). Call me maybe: Methods and practical implementation of artificial intelligence in call center arrivals' forecasting. *Journal of Business Research*, 123, 267–278.
- Allal-Chérif, O., Simón-Moya, V., & Ballester, A. C. C. (2021). Intelligent purchasing: How artificial intelligence can redefine the purchasing function. *Journal of Business Research*, 124, 69–76.
- Anand, G., & Ward, P. T. (2004). Fit, flexibility and performance in manufacturing: Coping with dynamic environments. *Production and Operations Management*, 13(4), 369–385.
- Armstrong, J. S., & Overton, T. S. (1977). Estimating nonresponse bias in mail surveys. *Journal of Marketing Research*, 14(3), 396–402.
- Asatiani, A., Malo, P., Nagbol, P. R., Penttinen, E., Rinta-Kahila, T., & Salovaara, A. (2020). Challenges of explaining the behavior of black-box AI systems. *MIS Quarterly Executive*, 19(4), 259–278.
- Barua, A., Konana, P., Whinston, A. B., & Yin, F. (2004). An empirical investigation of net-enabled business value. *MIS Quarterly*, 28(4), 585–620.
- Battleson, D. A., West, B. C., Kim, J., Ramesh, B., & Robinson, P. S. (2016). Achieving dynamic capabilities with cloud computing: An empirical investigation. *European Journal of Information Systems*, 25(3), 209–230.
- Benbya, H., Davenport, T. H., & Pachidi, S. (2020). Artificial intelligence in organizations: Current state and future opportunities. *MIS Quarterly Executive*, 19(4), ix–xxi.
- Benbya, H., Pachidi, S., & Jarvenpaa, S. (2021). Special issue editorial: Artificial intelligence in organizations: Implications for information systems research. *Journal of the Association for Information Systems*, 22(2), 281–303.
- Benner, M. J., & Tushman, M. L. (2015). Reflections on the 2013 Decade Award—"Exploitation, exploration, and process management: The productivity dilemma revisited" ten years later. *Academy of Management Review*, 40(4), 497–514.
- Berente, N., Gu, B., Recker, J., & Santhanam, R. (2021). Managing artificial intelligence. *MIS Quarterly*, 45(3), 1433–1450.
- Bhattacharjee, A., & Hikmet, N. (2007). Physicians' resistance toward healthcare information technology: A theoretical model and empirical test. *European Journal of Information Systems*, 16(6), 725–737.
- Bhoovaraghavan, S., Vasudevan, A., & Chandran, R. (1996). Resolving the process vs. product innovation dilemma: A consumer choice theoretic approach. *Management Science*, 42(2), 232–246.
- Bourgeois, L., & Eisenhardt, K. M. (1988). Strategic decision processes in high velocity environments: Four cases in the microcomputer industry. *Management Science*, 34(7), 816–835.
- Bradley, R. V., Pratt, R. M., Byrd, T. A., Outlay, C. N., & Wynn, D. E., Jr (2012). Enterprise architecture, IT effectiveness and the mediating role of IT alignment in US hospitals. *Information Systems Journal*, 22(2), 97–127.
- Brock, J. K. U., & Von Wangenheim, F. (2019). Demystifying AI: What digital transformation leaders can teach you about realistic artificial intelligence. *California Management Review*, 61(4), 110–134.
- Butner, K., Ivory, T., Albertoni, M., & Sotharan, K. (2020). *Automation and the future of work*. From: IBM Research Insights. <https://www.ibm.com/downloads/cas/O17AWNv6>.
- Capatina, A., Kachour, M., Lichy, J., Micu, A., Micu, A. E., & Codignola, F. (2020). Matching the future capabilities of an artificial intelligence-based software for social media marketing with potential users' expectations. *Technological Forecasting and Social Change*, 151, Article 119794.
- Chakravarty, A., Grewal, R., & Sambamurthy, V. (2013). Information technology competencies, organizational agility, and firm performance: Enabling and facilitating roles. *Information Systems Research*, 24(4), 976–997.
- Chatterjee, L., Feng, C., Nakata, C., & Sivakumar, K. (2023). The environmental turbulence concept in marketing: A look back and a look ahead. *Journal of Business Research*, 161, Article 113775.
- Cheatham, B., Cosmas, A., Mehta, N., & Shah, D. (2020). *How to build AI with (and for) everyone in your organization*. Retrieved From: McKinsey & Company. <https://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/how-to-build-ai-with-and-for-everyone-in-your-organization>.
- Chen, D. Q., Preston, D. S., & Swink, M. (2021). How big data analytics affects supply chain decision-making: An empirical analysis. *Journal of the Association for Information Systems*, 22(5), 1224–1244.
- Chen, Y., Wang, Y., Nevo, S., Benitez, J., & Kou, G. (2017). Improving strategic flexibility with information technologies: Insights for firm performance in an emerging economy. *Journal of Information Technology*, 32(1), 10–25.
- Chen, Y., Wang, Y., Nevo, S., Jin, J., Wang, L., & Chow, W. S. (2014). IT capability and organizational performance: The roles of business process agility and environmental factors. *European Journal of Information Systems*, 23(3), 326–342.
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed). Hillsdale, NJ: Erlbaum.
- Côrte-Real, N., Ramalho, N., Oliveira, T., & Morgado, E. M. (2017). Digital business transformation and strategy: What do we know so far? A systematic literature review and research agenda. *International Journal of Information Management*, 37(6), 557–570.
- Damanpour, F. (1991). Organizational innovation: A meta-analysis of effects of determinants and moderators. *Academy of Management Journal*, 34, 555–590.
- Damanpour, F., & Gopalakrishnan, S. (2001). The dynamics of the adoption of product and process innovations in organizations. *Journal of Management Studies*, 38(1), 45–65.
- Davenport, T. H. (2018). From analytics to artificial intelligence. *Journal of Business Analytics*, 1(2), 73–80.
- Davenport, T. H., Guha, A., Grewal, D., & Bressgott, T. (2020). How artificial intelligence will change the future of marketing. *Journal of the Academy of Marketing Science*, 48(1), 24–42.
- Davenport, T. H., & Mahidhar, V. (2018). What's your cognitive strategy. *MIT Sloan Management Review*, 59(4), 19–23.
- Davenport, T. H., & Ronanki, R. (2018). Artificial intelligence for the real world. *Harvard Business Review*, 96(1), 108–116.
- Deloitte. (2020). *Automation with intelligence: Pursuing organization-wide reimagination*. Retrieved From: Deloitte Insights. https://www2.deloitte.com/content/dam/insights/us/articles/73699-global-intelligent-automation-survey/DI_Automation-with-intelligence.pdf.
- Dess, G. G., & Beard, D. W. (1984). Dimensions of organizational task environments. *Administrative Science Quarterly*, 29(1), 52–73.
- Doherty, N. F., & Terry, M. (2009). The role of IS capabilities in delivering sustainable improvements to competitive positioning. *Journal of Strategic Information Systems*, 18(2), 100–116.

- Duan, Y., Edwards, J. S., & Dwivedi, Y. K. (2019). Artificial intelligence for decision making in the era of Big Data—evolution, challenges, and research agenda. *International Journal of Information Management*, 48, 63–71.
- Eisenhardt, K. M., & Martin, J. A. (2000). Dynamic capabilities: What are they? *Strategic Management Journal*, 21(10–11), 1105–1121.
- Elazhary, M., Popović, A., de Souza, H., Bermejo, P., & Oliveira, T. (2023). How information technology governance influences organizational agility: The role of market turbulence. *Information Systems Management*, 40(2), 148–168.
- Fernández-Rovira, C., Valdés, J.Á., Molleví, G., & Nicolas-Sans, R. (2021). The digital transformation of business. Towards the datafication of the relationship with customers. *Technological Forecasting and Social Change*, 162, Article 120339.
- Fornell, C., & Larcker, D. F. (1981). Structural equation models with unobservable variables and measurement error: Algebra and statistics. *Journal of Marketing Research*, 18(3), 382–388.
- Frazier, P. A., Tix, A. P., & Barron, K. E. (2004). Testing moderator and mediator effects in counseling psychology research. *Journal of Counseling Psychology*, 51(1), 115.
- Georgsdottir, A. S., & Getz, I. (2004). How flexibility facilitates innovation and ways to manage it in organizations. *Creativity and Innovation Management*, 13(3), 166–175.
- Gregory, R. W., Henfridsson, O., Kaganer, E., & Kyriakou, H. (2021). The role of artificial intelligence and data network effects for creating user value. *Academy of Management Review*, 46(3), 534–551.
- Grover, V., Kohli, R., & Guo, C. (2018). Business value of IT in the digital era: A study of IT-enabled digital transformation. *Journal of Management Information Systems*, 35(3), 984–1015.
- Gupta, S., Modgil, S., Bhattacharyya, S., & Bose, I. (2022). Artificial intelligence for decision support systems in the field of operations research: Review and future scope of research. *Annals of Operations Research*, 308, 1–60.
- Haefner, N., Vincent, J., Parida, V., & Gassmann, O. (2021). Artificial intelligence and innovation management: A review, framework, and research agenda. *Technological Forecasting and Social Change*, 162, Article 120392.
- Helfat, C. E. (2022). Strategic organization, dynamic capabilities, and the external environment. *Strategic Organization*, 20(4), 734–742.
- Helfat, C. E., Finkelstein, S., Mitchell, W., Peteraf, M. A., Singh, H., Teece, D. J., & Winter, S. G. (2007). *Dynamic Capabilities: Understanding Strategic Change in Organizations*. Carlton: Blackwell.
- Helfat, C. E., & Peteraf, M. A. (2015). Managerial cognitive capabilities and the microfoundations of dynamic capabilities. *Strategic Management Journal*, 36(6), 831–850.
- Hisrich, R. D. (1990). Entrepreneurship/Intrapreneurship. *American Psychologist*, 45, 209–222.
- Hollebeek, L. D., Sprott, D. E., & Brady, M. K. (2021). Rise of the machines? Customer engagement in automated service interactions. *Journal of Service Research*, 24(1), 3–8.
- Holmqvist, M., & Pessi, K. (2006). Agility through scenario development and continuous implementation: A global aftermarket logistics case. *European Journal of Information Systems*, 15(2), 146–158.
- Hoonsoon, D., & Puriwat, W. (2019). Organizational agility: Key to the success of new product development. *IEEE Transactions on Engineering Management*, 68(6), 1722–1733.
- Huang, M. H., & Rust, R. T. (2021a). Engaged to a robot? The role of AI in service. *Journal of Service Research*, 24(1), 30–41.
- Huang, M. H., & Rust, R. T. (2021b). A strategic framework for artificial intelligence in marketing. *Journal of the Academy of Marketing Science*, 49(1), 30–50.
- Huang, M. H., & Rust, R. T. (2022). A framework for collaborative artificial intelligence in marketing. *Journal of Retailing*, 98(2), 209–223.
- Huang, M. H., Rust, R., & Maksimovic, V. (2019). The feeling economy: Managing in the next generation of artificial intelligence (AI). *California Management Review*, 61(4), 43–65.
- Hulland, J., Baumgartner, H., & Smith, K. M. (2018). Marketing survey research best practices: Evidence and recommendations from a review of JAMS articles. *Journal of the Academy of Marketing Science*, 46, 92–108.
- Iansati, M., & Lakhari, K. R. (2020). Competing in the age of AI. *Harvard Business Review*. January–February.
- Im, S., & Workman, J. P., Jr. (2004). Market orientation, creativity, and new product performance in high-technology firms. *Journal of Marketing*, 68(2), 114–132.
- Jaworski, B. J., & Kohli, A. K. (1993). Market orientation: Antecedents and consequences. *Journal of Marketing*, 57(3), 53–70.
- Jussupow, E., Spohrer, K., Heinzl, A., & Gawlitza, J. (2021). Augmenting medical diagnosis decisions? An investigation into physicians' decision-making process with artificial intelligence. *Information Systems Research*, 32(3), 713–735.
- Kalaigian, K., Tuli, K. R., Kushwaha, T., Lee, L., & Gal, D. (2021). Marketing agility: The concept, antecedents, and a research agenda. *Journal of Marketing*, 85(1), 35–58.
- Kaplan, A., & Haenlein, M. (2019). Siri, Siri, in my hand: Who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence. *Business Horizons*, 62(1), 15–25.
- Kathuria, A., Mann, A., Khuntia, J., Saldanha, T. J., & Kauffman, R. J. (2018). A strategic value appropriation path for cloud computing. *Journal of Management Information Systems*, 35(3), 740–775.
- Kiron, D., & Schrage, M. (2019). Strategy for and with AI. *MIT Sloan Management Review*, 60(4), 30–35.
- Kock, N. (2022). WarpPLS user manual: Version 8.0. *ScriptWarp Systems*: Laredo, TX, USA.
- Kohli, A. K., & Jaworski, B. J. (1990). Market orientation: The construct, research propositions, and managerial implications. *Journal of Marketing*, 54(2), 1–18.
- Kohli, R., & Grover, V. (2008). Business value of IT: An essay on expanding research directions to keep up with the times. *Journal of the Association for Information Systems*, 9(1), 23–39.
- Kulkarni, U., Robles-Flores, J. A., & Popović, A. (2017). Business intelligence capability: The effect of top management and the mediating roles of user participation and analytical decision making orientation. *Journal of the Association for Information Systems*, 18(7), 516–541.
- Kyläheiko, K., & Majanen, P. (2020). Industry 4.0. transformation challenge in light of dynamic capabilities. *Technical, Economic and Societal Effects of Manufacturing*, 4, 169–189.
- Lee, N. C. A., Wang, E. T., & Grover, V. (2020). IOS drivers of manufacturer-supplier flexibility and manufacturer agility. *Journal of Strategic Information Systems*, 29(1), Article 101594.
- Lee, O. K., Sambamurthy, V., Lim, K. H., & Wei, K. K. (2015). How does IT ambidexterity impact organizational agility? *Information Systems Research*, 26(2), 398–417.
- Lin, C., & Kunnathur, A. (2019). Strategic orientations, developmental culture, and big data capability. *Journal of Business Research*, 105, 49–60.
- Lindell, M. K., & Whitney, D. J. (2001). Accounting for common method variance in cross-sectional research designs. *Journal of Applied Psychology*, 86(1), 114.
- Lu, Y., & Ramamurthy, K. (2011). Understanding the link between information technology capability and organizational agility: An empirical examination. *MIS Quarterly*, 35(4), 931–954.
- Luo, X., Tong, S., Fang, Z., & Qu, Z. (2019). Frontiers: Machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases. *Marketing Science*, 38(6), 937–947.
- Makarius, E. E., Mukherjee, D., Fox, J. D., & Fox, A. K. (2020). Rising with the machines: A sociotechnical framework for bringing artificial intelligence into the organization. *Journal of Business Research*, 120, 262–273.
- Malhotra, N. K., Kim, S. S., & Patil, A. (2006). Common method variance in IS research: A comparison of alternative approaches and a reanalysis of past research. *Management Science*, 52(12), 1865–1883.
- Manis, K. T., & Madhavaram, S. (2023). AI-Enabled marketing capabilities and the hierarchy of capabilities: Conceptualization, proposition development, and research avenues. *Journal of Business Research*, 157, Article 113485.
- Mathieson, K., Peacock, E., & Chin, W. W. (2001). *Extending the technology acceptance model: The influence of perceived user resources*. *ACM SIGMIS Database: the DATABASE for Advances in Information Systems*, 32(3), 86–112.
- Mellahi, K. (2001). Differences and similarities in future managerial values: A five cultures comparative study. *Cross Cultural Management: An International Journal*, 8(1), 45–58.
- Melville, N., Kraemer, K., & Gurbaxani, V. (2004). Information technology and organizational performance: An integrative model of IT business value. *MIS Quarterly*, 28(2), 283–322.
- Mitchell, J. R., Shepherd, D. A., & Sharfman, M. P. (2011). Erratic strategic decisions: When and why managers are inconsistent in strategic decision making. *Strategic Management Journal*, 32(7), 683–704.
- Moore, G. C., & Benbasat, I. (1991). Development of an instrument to measure the perceptions of adopting an information technology innovation. *Information Systems Research*, 2(3), 192–222.
- Morgan, N. A., Vorhies, D. W., & Mason, C. H. (2009). Market orientation, marketing capabilities, and firm performance. *Strategic Management Journal*, 30(8), 909–920.
- Mumford, M. D., & Gustafson, S. B. (1988). Creativity syndrome: Integration, application and innovation. *Psychological Bulletin*, 103, 27–43.
- Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital innovation management: Reinventing innovation management research in a digital world. *MIS Quarterly*, 41(1), 223–238.
- Narayandas, D., & Sengupta, A. (2023). Using AI to adjust your marketing and sales in a volatile world. *Harvard Business Review*. April 12. Retrieved from <https://hbr.org/2023/04/using-ai-to-adjust-your-marketing-and-sales-in-a-volatile-world>.
- Nazir, S., & Pinsonneault, A. (2012). IT and firm agility: An electronic integration perspective. *Journal of the Association for Information Systems*, 13(3), 150–171.
- Ngai, E. W., Chau, D. C., & Chan, T. L. A. (2011). Information technology, operational, and management competencies for supply chain agility: Findings from case studies. *Journal of Strategic Information Systems*, 20(3), 232–249.
- Oke, A., Burke, G., & Myers, A. (2007). Innovation types and performance in growing UK SMEs. *International Journal of Operations & Production Management*, 27(7), 735–753.
- Organization for Economic Co-operation and Development (OECD). (2019). ICT access and usage by households and individuals. From: https://stats.oecd.org/Index.aspx?DataSetCode=ICT_HH2.
- Özsomer, A., Calantone, R. J., & Di Bonetto, A. (1997). What makes firms more innovative? A look at organizational and environmental factors. *Journal of Business & Industrial Marketing*, 12(6), 400–416.
- Paschen, U., Pitt, C., & Kietzmann, J. (2020). Artificial intelligence: Building blocks and an innovation typology. *Business Horizons*, 63(2), 147–155.
- Pavlou, P. A., & El Sawy, O. A. (2006). From IT leveraging competence to competitive advantage in turbulent environments: The case of new product development. *Information Systems Research*, 17(3), 198–227.
- Pavlou, P. A., & El Sawy, O. A. (2011). Understanding the elusive black box of dynamic capabilities. *Decision Sciences*, 42(1), 239–273.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879–903.
- Polites, G. L., & Karahanna, E. (2012). Shackled to the status quo: The inhibiting effects of incumbent system habit, switching costs, and inertia on new system acceptance. *MIS Quarterly*, 36(1), 21–42.

- Prajogo, D. I., & Ahmed, P. K. (2006). Relationships between innovation stimulus, innovation capacity, and innovation performance. *R&D Management*, 36(5), 499–515.
- Prange, C. (2021). Agility as the discovery of slowness. *California Management Review*, 63(4), 27–51.
- Puriwat, W., & Hoonsopon, D. (2021). Cultivating product innovation performance through creativity: The impact of organizational agility and flexibility under technological turbulence. *Journal of Manufacturing Technology Management*, 33(4), 741–762.
- Queiroz, M., Tallon, P. P., Sharma, R., & Coltman, T. (2018). The role of IT application orchestration capability in improving agility and performance. *Journal of Strategic Information Systems*, 27(1), 4–21.
- Raju, P., & Koch, R. (2019). Can RPA improve agility? *Strategic Finance* (March), 68–69.
- Ravichandran, T. (2018). Exploring the relationships between IT competence, innovation capacity and organizational agility. *Journal of Strategic Information Systems*, 27(1), 22–42.
- Ravichandran, T., Lertwongsatien, C., & Lertwongsatien, C. (2005). Effect of information systems resources and capabilities on firm performance: A resource-based perspective. *Journal of Management Information Systems*, 21(4), 237–276.
- Rosenblat, A. (2018). *Uberland: How Algorithms are Rewriting the Rules of Work*. Univ of California Press.
- Sambamurthy, V., Bharadwaj, A., & Grover, V. (2003). Shaping agility through digital options: Reconceptualizing the role of information technology in contemporary firms. *MIS Quarterly*, 27(2), 237–263.
- Shrestha, Y. R., Ben-Menahem, S. M., & Von Krogh, G. (2019). Organizational decision-making structures in the age of artificial intelligence. *California Management Review*, 61(4), 66–83.
- Slevin, D. P., & Covin, J. G. (1997). Strategy formation patterns, performance, and the significance of context. *Journal of Management*, 23(2), 189–209.
- Straub, D. W. (1989). Validating instruments in MIS research. *MIS Quarterly*, 13(2), 147–169.
- Tallon, P. P., & Pinsonneault, A. (2011). Competing perspectives on the link between strategic information technology alignment and organizational agility: Insights from a mediation model. *MIS Quarterly*, 35(2), 463–486.
- Tan, T. F., Netessine, S., & Hitt, L. (2017). Is Tom Cruise threatened? An empirical study of the impact of product variety on demand concentration. *Information Systems Research*, 28(3), 643–660.
- Tanriverdi, H., & Venkatraman, N. (2005). Knowledge relatedness and the performance of multibusiness firms. *Strategic Management Journal*, 26(2), 97–119.
- Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, 28(13), 1319–1350.
- Teece, D. J. (2014). The foundations of enterprise performance: Dynamic and ordinary capabilities in an (economic) theory of firms. *Academy of Management Perspectives*, 28(4), 328–352.
- Teece, D., Peteraf, M., & Leih, S. (2016). Dynamic capabilities and organizational agility: Risk, uncertainty, and strategy in the innovation economy. *California Management Review*, 58(4), 13–35.
- Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7), 509–533.
- Trantopoulos, K., von Krogh, G., Wallin, M. W., & Woerter, M. (2017). External knowledge and information technology. *MIS Quarterly*, 41(1), 287–300.
- Tushman, M. L., & Nadler, D. (1986). Organizing for innovation. *California Management Review*, 28(3), 74–92.
- Tussyadiah, I. (2020). A review of research into automation in tourism: Launching the annals of tourism research curated collection on artificial intelligence and robotics in tourism. *Annals of Tourism Research*, 81, Article 102883.
- Wang, C. L., & Ahmed, P. K. (2004). The development and validation of the organizational innovativeness construct using confirmatory factor analysis. *European Journal of Innovation Management*, 7(4), 303–313.
- Wang, W. T., Chih, W. H., & Jiang, J. J. (2012). Analyzing the determinants of business value creation in e-business settings. *Information & Management*, 49(8), 385–397.
- Whittlestone, J., Nyrop, R., Alexandrova, A., Dihal, K., & Cave, S. (2019). *Ethical and Societal Implications of Algorithms, Data, and Artificial Intelligence: A Roadmap for Research*. London: Nuffield Foundation.
- Wilden, R., & Gudergan, S. P. (2015). The impact of dynamic capabilities on operational marketing and technological capabilities: Investigating the role of environmental turbulence. *Journal of the Academy of Marketing Science*, 43, 181–199.
- Zahra, S. A., & Garvis, D. M. (2000). International corporate entrepreneurship and firm performance: The moderating effect of international environmental hostility. *Journal of Business Venturing*, 15(5–6), 469–492.
- Zhou, K. Z., & Wu, F. (2010). Technological capability, strategic flexibility, and product innovation. *Strategic Management Journal*, 31(5), 547–561.

Yulia Sullivan, Ph.D. is an Associate Professor of Information Systems and Business Analytics in Hankamer Business School at Baylor University. She earned her Ph.D. degree in information systems from the University of North Texas. Her research interests include cognitive information systems, the value of IT for organizations, artificial intelligence & ethics, human-computer interaction, and research methods. Her work has been published in various journals, including *MIS Quarterly*, *Journal of the Association for Information Systems*, *Journal of Business Ethics*, *International Journal of Information Management*, *Journal of Medical Internet Research*, and others. She has also presented her research at many international conferences, including the International Conference on Information Systems and the Hawaii International Conference on System Sciences. She is also a co-author of *Conducting Mixed-Methods Research: From Classical Social Sciences to the Age of Big Data and Analytics*.

Samuel Fosso Wamba, Ph. D. is a Professor at TBS Education, France, and a Distinguished Visiting Professor at The University of Johannesburg, South Africa. He earned his Ph.D. in industrial engineering at the Polytechnic School of Montreal, Canada. He earned his Ph.D. in industrial engineering at the Polytechnic School of Montreal, Canada. He also holds two master's degrees: one MSc in mathematics from the University of Sherbrooke in Canada and another MSc in electronic commerce with a major in computer science from HEC Montreal, Canada. His current research focuses on the business value of I.T., inter-organizational systems adoption, use and impacts, supply chain management, electronic commerce, blockchain, artificial intelligence for business, social media, business analytics, big data, and open data. He is a CompTIA RFID+ Certified Professional and the Academic Co-Founder of RFID Academia. Apart from teaching and conducting research, He leads the newly created Center of Excellence in Artificial Intelligence & Business Analytics at TBS Education. He has been recently recognized as the most influential scholar in big data analytics and enterprises based on the number of published articles and citations and among the 2% of the most influential scholars in the world based on the Mendeley database that includes 100,000 top scientists. He ranks in ClarivateTM's 1% most cited scholars in the world for 2020, from the "Highly Cited ResearchersTM" list that identifies global research scientists who show exceptional influence-reflected in the publication of multiple papers frequently cited by their peers. He ranks in CDO Magazine's Leading Academic Data Leaders 2021.